

STRATEGIC ORIENTATION REPORT

THE ECONOMIC HEAVENLY WAY

Diagnosing Chinese Stasis and Deploying the Territorial Regeneration Rent (TRR)

Strategic report for policymakers, senior NDRC officials and geopolitical strategists

Author: Bruno DE DOMINICIS (*Independent Think Tank*)

Date: September 2026 **DOI:** 10.5281/zenodo.xxxxx

DISCLAIMER AND SCIENTIFIC TRACEABILITY

This document is a strategic summary intended to guide the operational management of the 15th Five-Year Plan. The mathematical framework (Clifford algebra $Cl(6, 6)$ modeling, Λ_{72} lattice), the empirical validations, and the application of the Rent Extraction Theory to Chinese economic history are covered in a 250-page document available on Zenodo and GitHub.

Contents

EXECUTIVE SUMMARY	6
FOREWORD – GENESIS AND EPISTEMOLOGY OF THE MODEL	7
The structural isomorphism: the economy as a discrete information system	7
The universal invariant $64 \rightarrow 20$: interdisciplinary convergence	7
Extension to $Cl(6,6)$: modeling economic dualities	8
From complex systems theory to an institutional steering tool	8
PART I – THE DIAGNOSIS: WHY CHINA IS IN STASIS	9
Introduction	9
Chapter 1 – The exhaustion of the land-based engine (2018-2026)	9
1.1. Macroeconomic data 2026: collapse of construction, LGFV debt, youth unemployment	9
1.2. The Nodal Point: end of the material frontier and saturation of space	9
1.3. Sovereign Wealth destruction (Evergrande, Three Red Lines) as an adjustment mechanism	10
Chapter 2 – The dead-end of the immaterial and temporal avoidance	10
2.1. Carbon, Living, Cognitive Data: three bubbles without spatial thickness	10
2.2. The SECC (Computing Capacity Export Strategy): an external palliative facing overproduction	11
2.3. The mass employment deficit: elasticity 0.14 against 1.8 for construction	11
Chapter 3 – The phenomenon of “Tang Ping” as a social symptom	11
3.1. Withdrawal from competition: sociological data and testimonies	11
3.2. Stasis as an absorbing attractor and risk of long depression	12
PART II – THE TOPOLOGICAL ENGINE: GENERAL PRESENTATION	12
Introduction	12
Chapter 4 – Fundamental principles of modeling	12
4.1. The economy as a discrete and dynamic information system	12
4.2. The universal invariant $64 \rightarrow 20$: applicability to economic regimes	13
4.3. The 6 binary variables (Inflation, Purchasing power, Profits, Extraction, Dispersion, Land regime)	13
Chapter 5 – Mapping the 20 dynamic regimes: beyond the Kondratiev cycle	13
5.1. The pentad triplet as the identity of an economic regime	13
5.2. The polarity gradient $3P \rightarrow 3N$ (from polarized growth to depression)	14
5.3. The 7 spectral thresholds S_1 to S_7 : the energy gates of the system	14
Chapter 6 – The 4 global observables of Stasis	14
6.1. η (asymmetry): the Qi of the system – expansion or recession	15
6.2. d (spectral dimension): harmony – diversity or concentration	15
6.3. gap (fragility): the margin before collapse	15
6.4. R_{seuil} (tension on thresholds): systemic red alert	15
6.5. Visual Synthesis: The Spectral Footprint	16
PART III – THE ADMINISTRATIVE ARCHITECTURE OF THE MODEL	17
Introduction	17
Chapter 7 – A steering grid nested within the State organization	17
7.1. Resonance principle: granularity of diagnosis = granularity of intervention	17
7.2. The 12 spectral leaves (meridians) and their correspondence with economic sectors	17
Chapter 8 – Level 1: Strategic steering (Beijing)	18

8.1. The 4 global observables for the State Council	18
8.2. Decision rules: when to activate the TRR, when to reform the currency	18
8.3. The composite Stasis index $\mathcal{F}_{Stase}$: the quantified “Mandate of Heaven”	18
Chapter 9 – Level 2: Tactical steering (Provinces)	19
9.1. The 48 observables per spectral leaf ($\eta_f, d_f, gap_f, R_{seuil,f}$)	19
9.2. Adjustment of GEPs (Gross Ecosystem Product) and CURs by region	19
9.3. Case of Zhejiang: imbalance between Innovation (e_1) and Accumulation (f_3)	19
Chapter 10 – Level 3: Operational steering (Counties / Pilot zones)	19
10.1. The 144 projected pentads: the economy in “acupuncture points”	19
10.2. Activity, polarity, connectivity, frustration – the 4 signs of the pentad	20
10.3. Case of Lishui: desaturation of node $N_5^{(f_3)}$ by the Two Mountains Bank	20
Chapter 11 – Macro-systemic governance: the National Observatory dashboard	20
11.1. The 7 spectral thresholds as an early warning system	20
11.2. Wuxing tensions: rebalancing the 5 phases (Wood, Fire, Earth, Metal, Water)	20
11.3. The inter-leaf correlation matrix: detecting regional fragmentation	21
PART IV – THE TRR AS A TRANSITION OPERATOR	21
Introduction	21
Chapter 12 – Formalization of the Territorial Regeneration Rent (TRR)	22
12.1. The TRR as a controlled crossing through the threshold N_4	22
12.2. The operator T_{mixed} : combining $T_{structure}, T_{fire}, T_{water}$	22
12.3. The 3 structural reforms as components of the engine	22
Chapter 13 – Pilot zones: the TRR already at work	22
13.1. Lishui (Zhejiang): Two Mountains Bank and GEP securitization	23
13.2. Sanming / Nanping (Fujian): Lin-piao and regenerated forests	23
13.3. Xingguo / Ganzhou (Jiangxi): watersheds and contaminated soils	23
13.4. Hainan: island GEP and prohibition of speculation	23
Chapter 14 – Deployment schedule 2026-2035	23
14.1. Pilot phase (2024-2025): crossing S_5 (Fire)	23
14.2. Provincial deployment (2026-2028): crossing S_6 (Water)	23
14.3. Acceleration (2029-2030): crossing S_7 (Octave jump)	24
14.4. Generalization (2031-2035): stabilization on C_P (new Extraction cycle)	24
Chapter 15 – Simulation of the two scenarios	24
15.1. Scenario 1: Without TRR – retropolarity and tenth decline (observables 2032)	24
15.2. Scenario 2: With TRR – exit from Stasis and healthy growth (observables 2032)	24
15.3. Synthetic comparison of indicators	25
PART V – STRATEGIC DASHBOARD	25
Introduction	25
Chapter 16 – The multi-level dashboard (China 2026)	25
16.1. Level 1 – Beijing Radar	27
16.2. Level 2 – Heat map of the 12 spectral leaves	27
16.3. Level 3 – Threshold vector	27
16.4. Level 4 – Active pentads	28
16.5. Level 5 – Wuxing tensions	28

Chapter 17 – Traffic light system and emergency procedures	28
17.1. Alert thresholds by administrative level	28
17.2. Intervention protocol in the red zone	28
17.3. Monitoring indicators for the 15th Five-Year Plan (2026-2030)	29
PART VI – STRATEGIC AND GEOPOLITICAL IMPLICATIONS	29
Introduction	29
Chapter 18 – The TRR and the preservation of the Mandate of Heaven	30
18.1. The Mandate as an implicit social contract	30
18.2. The TRR as a restoration of political virtue (德, De)	30
18.3. Response to the demographic crisis: mass employment and living territories	30
Chapter 19 – Cultural correspondences as a framework for legitimization	30
19.1. Yi Jing: 64 hexagrams, 64 configurations – the modernized divinatory dialogue	30
19.2. Wuxing: the 5 phases as regulation of Sheng/Ke cycles	31
19.3. Traditional Chinese Medicine: 12 meridians, 4 examinations	31
Chapter 20 – The geopolitical impact of the TRR: the end of spatial avoidance	32
20.1. Reduction of dependence on exports: the TRR as a substitute for economic imperialism	32
20.2. Redefinition of the New Silk Roads: from spatial expansion to the diplomacy of regeneration	32
20.3. Sino-American rivalry: the asymmetric advantage of endogenous regeneration over Western financialization	32
GENERAL CONCLUSION	33
APPENDICES (Strategic Selection)	35
Appendix A – Glossary of 15 technical terms (decision-maker)	35
Appendix B – The 20 attractors	36
Appendix C – The administrative hierarchy of the model	39
Appendix D – The 7 spectral thresholds and their political significance	39
Appendix E – Mapping of TRR pilot zones	40
Appendix F – Detailed simulation of the two scenarios (with/without TRR)	41
Appendix G – The 9 Chinese dynastic declines	42
Appendix H – Genesis of the model: from Clifford algebras to complexity invariants (technical version)	46
1. From Rent Extraction Theory to the discretization of the economy	46
2. The algebra $Cl(6, 0)$ and filtration by nilpotence (Rowlands / De Dominicis)	46
3. Extension to $Cl(6, 6)$: Sheng/Ke duality and Λ_{72} lattice	46
Appendix I – Dashboard user manual (for technical teams)	47
1. Functional architecture and data ingestion	47
2. Observable calculation algorithm	47
3. Alert and validation procedure (Example: China 2026 Red Alert)	48
TECHNICAL ANNEX DOCUMENT	49
Mathematical Foundations and Architecture of the $Cl(6,6)$ Steering System	49
1. Basic Topological Architecture: From $Cl(6,0)$ to $Cl(6,6)$	49
2. The Exceptional Lattice Λ_{72} and Spectral Thresholds	50
3. Dirac Operator and Calculation of Global Observables (Level 1)	51
4. Hierarchical Extension and Composite Index (Levels 2 to 5)	52
5. Algorithmic Implementation Note	53

REFERENCES AND KEY SOURCES	53
1. Theoretical and topological foundations	54
2. Macroeconomic and sectoral data (China 2026)	54
3. Institutional and geopolitical framework	54
THEORETICAL AND MATHEMATICAL ANNEX DOCUMENT	55
Foundations, Isomorphisms and Reference Tables of the Cl(6,6) Model	55
PART A: ADVANCED MATHEMATICAL FOUNDATIONS	55
1. The Exceptional Lattice Λ_{72} and the Universal Constants β_k	55
2. The Transition Operator T and its Selection Rules	56
PART B: EXHAUSTIVE REFERENCE TABLES	56
Complete Table of the 144 Economic Pentads of Cl(6,6)	56
4. Comparison of Post-Nodal Point Alternatives	66
PART C: EPISTEMOLOGY AND STRUCTURAL ISOMORPHISMS	66
5. Vandermeersch's Contribution: Chinese Divinatory Reason	66
6. Philosophical Synthesis: The spectral resolution of the Xiong Shili – Nāgārjuna controversy	67
7. The Aporias of Neoclassical and Marxist Economic Theories	67
8. Isomorphism with Tang Alliances (Wufu) and Mandarin Phonology	68

EXECUTIVE SUMMARY

This strategic document formalizes a paradigm shift in the understanding and steering of the Chinese economy. Inspired by the Rent Extraction Theory (TdR) and formalized by the Clifford algebra $Cl(6,6)$, it proposes a discrete topological model that reveals three counterintuitive facts about the current crisis:

1. **The crisis is not an anomaly, but a law of conservation:** The real estate collapse and the local government debt (LGFV) crisis are not governance failures, but the inevitable attainment of the “Nodal Point” — the physical exhaustion of the stock of land-based Value available for extraction.
2. **The immaterial is a morphological dead-end:** The transition towards Artificial Intelligence and carbon (the “Third Mutation”) does not create the necessary mass employment. With an employment elasticity of 0.14 (compared to 1.8 for construction), the immaterial generates an Algorithmic Stasis.
3. **Wealth destruction can be avoided:** For the first time in two millennia, a Chinese state has a peaceful tool — the Territorial Regeneration Rent (TRR) — to short-circuit the fatal cycle of dynastic collapses.
4. **The TRR as the foundation of a new diplomacy — The Diplomacy of Regeneration:** Beyond China, the TRR grounds a new diplomacy. By demonstrating that it is possible to exit the Nodal Point without massive wealth destruction, China offers the Global South a structural alternative to Western imperialism. Instead of exporting an extraction model (logistical colonization, factories vulnerable to the Sovereign Rent of host states), China can export a topological regeneration model: the Two Mountains Bank, the GEP in KPIs, the Regenerative Usufruct Certificates. This “Two Mountains diplomacy” positions the Economic Heavenly Way (Tiandao Jingji, 天道经济) as a viable new civilizational standard, an alternative to the exhausted Western model.

The 4-level steering architecture To operationalize this exit from the crisis, the model deploys an administrative reading grid armed with 4 global spectral observables (η, d, gap, R_{seuil}):

Table 1: 4-level steering architecture

Level	Scale	Observables	Typical Action	Horizon
Level 1	Beijing (Strategic)	4 global (η, d, gap, R_{seuil})	Activate TRR if $R_{seuil} > 0.7$	5–10 years
Level 2	Province (Tactical)	48 per spectral leaf	Adjust GEP, reorient CUR	1–3 years
Level 3	County (Operational)	144 projected pentads	Target CUR on active pentads	6–18 months
Level 4	Observatory (Alert)	7 thresholds + 66 relational	Detect pre-retropolarity	Real-time

The TRR as the only authorized trajectory Topologically, exiting Stasis requires the controlled crossing of the threshold N_4 . The TRR is the operator T_{mixed} that activates this passage. It combines three structural components (Wood, Earth, Metal) as well as Fire (shocks/innovation) and Water (money/credit), synthesized in 3 reforms:

- **Two Mountains Bank (T_{water}):** Securitization of the GEP to circumvent LGFV debt.

- **Integration of the GEP into KPIs** ($T_{structure}$): Reform of political incentives.
- **Regenerative Usufruct Certificate** (T_{fire}): Creation of a new value distribution cycle.

The spectral dashboard This model offers the Party-State an unprecedented governance tool, modeled on the four examinations of Traditional Chinese Medicine (望闻问切):

- **4 vital signs:** Asymmetry (η , the Qi), dimension (d , harmony), gap (fragility), threshold (R_{seuil} , tension).
 - **12 meridians:** The 12 sectoral spectral leaves (Innovation, Profits, Land, etc.).
 - **144 acupuncture points:** The 144 projected economic pentads, enabling fine territorial targeting to unblock blockage nodes.
-

FOREWORD – GENESIS AND EPISTEMOLOGY OF THE MODEL

Conventional macroeconomics (DSGE models) treats the economy as a continuous space of variables mechanically adjusting towards equilibrium. This approach structurally fails to anticipate abrupt ruptures (1929, 2008, Chinese Stasis) because it ignores the discrete, informational, and threshold-based nature of complex systems. The topological model $Cl(6,6)$ offers a radical epistemological alternative.

The structural isomorphism: the economy as a discrete information system

The Rent Extraction Theory (TdR) postulates that economic dynamics are not a continuous fluid, but a network of discrete configurations. Just as quantum physics, it models states and phase transitions. The mathematical tool of this discretization is the Clifford algebra $Cl(6, 0)$, which generates a space of 64 possible economic configurations, defined by 6 binary variables (inflation, purchasing power, profits, extraction, dispersion, land regime). The economy thus obeys the same topological laws as particle physics.

The universal invariant $64 \rightarrow 20$: interdisciplinary convergence

The major innovation of this model is the demonstration that the space of 64 configurations reduces, by topological filtration (nilpotence), to **20 stable attractors**. This invariant $64 \rightarrow 20$ is not speculation: it is universally verified in complex systems.

Table 2: The five manifestations of the invariant $64 \rightarrow 20$

Domain	Input (64)	Output (20)	Filtration
Particle physics	64 spin configurations	20 stable states	$Cl(6, 0)$ algebra / Λ_{72} lattice
Genetic code	64 codons	20 amino acids	Neighborhood rule (De Dominicis)
Mandarin phonology	64 combinations	20 phonological classes	Articulatory constraints
Alliances (Tang dynasty)	64 configurations	20 alliance attractors	Wufu (kinship) filtration
Economics (TdR)	64 configurations	20 attractors	Nilpotence

This convergence proves that correlative Chinese thought (*Yi Jing* and its 64 hexagrams) was not merely a divinatory metaphor, but a proto-empirical model of the topology of complex systems, of which $Cl(6, 0)$ algebra is the modern formalization.

Extension to $Cl(6,6)$: modeling economic dualities

If $Cl(6, 0)$ describes states, the extension to $Cl(6, 6)$ describes dynamics. It introduces 6 “anti-cosmic” variables (regulation, contraction) added to the 6 “cosmic” variables (expansion). This signature $(6, 6)$ perfectly models fundamental economic dualities, algebraic equivalents of the *Sheng* (generative) and *Ke* (regulatory) cycles of *Wuxing*. The space generates **144 pentads**, offering an unprecedented resolution of inter-sectoral tensions.

From complex systems theory to an institutional steering tool

This document is the synthesis of a decade of interdisciplinary research bridging structural geography (TdR), Clifford algebra (Rowlands), and complex network arithmetic (Nebe, Λ_{72}). Its objective is not theoretical, but eminently operational. By providing the State Council with a spectral dashboard at 5 levels of resolution, the model transforms the art of governing. It offers the Party-State the capacity to quantify its “Mandate of Heaven” (via the Stasis Index $\mathcal{F}_{Stase}$), to topographically locate blockage nodes (economic acupuncture), and to anticipate tipping points (retropolarity) with mathematical precision, paving the way for a physics of regulation.

This strategic working document constitutes a policy-oriented synthesis of a comprehensive scientific and technical report of 228 pages. To preserve the readability and operability of the text for decision-makers, the exhaustive mathematical demonstrations (diagonalization of the Dirac operator, arithmetic of the Λ_{72} lattice, full formalization of the 144 pentads), as well as the complete historiography of the 9 dynastic declines, have been condensed or relegated to appendices. The full report, intended for econometricians, mathematicians, and epistemologists, is available upon request and provides the formal proofs underlying the steering model presented here.

PART I – THE DIAGNOSIS: WHY CHINA IS IN STASIS

Introduction

The structural crisis that China is going through today — collapse of the real estate market, suffocating debt of municipalities, halt of mega-infrastructures — is neither a cyclical anomaly nor an ideological failure. It is the predictable stall of a growth engine that has simply exhausted its energy. For the public decision-maker, understanding this crisis requires stepping outside the neoclassical framework (which fails to model ruptures) and adopting the grid of the Rent Extraction Theory (TdR): the economy is modeled there as a discrete information system, governed by critical rupture thresholds. What China has been experiencing since 2018 is nothing other than the attainment of the **terminal Nodal Point**, the topological lock that marks the end of a centuries-old trend of material growth.

Chapter 1 – The exhaustion of the land-based engine (2018-2026)

1.1. Macroeconomic data 2026: collapse of construction, LGFV debt, youth unemployment

The diagnosis of Chinese Stasis is based on implacable macroeconomic indicators that reveal the abrupt halt of the engine of “Land Finance” (*Tudi Caizheng*). Data published in the first half of 2026 by the National Bureau of Statistics (NBS) and the IMF confirm the scale of the shock:

- **Collapse of the real estate sector:** New housing starts have fallen 68% compared to the 2021 peak. Real estate investment continues its contraction of –7.9% year-on-year in H1 2026, despite three successive stimulus plans by the Party-State. The real estate and construction sector, which represented ~25 to 30% of GDP, destroyed **more than 19 million direct and indirect jobs** between 2022 and June 2026.
- **The local debt impasse (LGFV):** Local Government Financing Vehicles (LGFVs), instruments of off-balance-sheet debt backed by the future value of land, are asphyxiated. The IMF estimates this debt at about 60,000 billion yuan (48% of GDP). Local governments are captive to a contradiction: they cannot let the system collapse (risk of systemic contagion), nor maintain it as is (Japan-style zombification) without recreating Value.
- **Youth unemployment:** The destruction of industrial and real estate jobs has hit young people hard. The real unemployment rate for 16-24 year-olds, including underemployment, now exceeds **60%**, a level comparable to the American Great Depression.

1.2. The Nodal Point: end of the material frontier and saturation of space

The Rent Extraction Theory postulates that growth is a phenomenon of structural disequilibrium: it rests on the transfer of “dormant” assets (land, labor) from the Space of Value to the Space of Wealth, financed by borrowing. This dynamic has an absolute limit, the **Nodal Point**, reached when the stock of easily mobilizable Value is exhausted.

From 1978 to 2018, China mobilized its immense stock of latent Value (maintained by the *Hukou*) at an unprecedented speed. To postpone the Nodal Point, the Party-State deployed spatial avoidance strategies by “extracting”

physical geography (Three Gorges Dam, South-North Water Transfer, West-East Gas Pipeline). However, around 2015, the urbanization rate exceeding 65%, spatial saturation was reached: the lands with strong externalities around coastal metropolises were fully absorbed, and Western mega-infrastructures reached their profitability limits (encountering “refractory Values” whose extraction cost exceeds the value produced). The topological wall was reached.

1.3. Sovereign Wealth destruction (Evergrande, Three Red Lines) as an adjustment mechanism

The conservation law of the TdR stipulates that at the Nodal Point, a destruction of Wealth is mandatory to recreate the conditions for a new cycle. Unlike the West, which socialized banking losses during the 2008 crisis (chaotic destruction by the market), the Party-State chose **sovereign selective demolition**.

The “Three Red Lines” policy (2020) and the Evergrande precedent illustrate this adjustment mechanism. By allowing the developer to default on its *offshore* debt while imposing the slogan *Baojiaolou* (“Guarantee the delivery of housing”), the State operated a triple adjustment:

1. **Sanction of rentier drift:** destruction of shareholder and speculative financial Wealth.
2. **Geopolitical firebreak:** cutting the channel of contagion from Western financial speculation, immunizing the domestic banking system.
3. **Preservation of the social contract:** protection of use-Wealth (the concrete territorial anchoring of the family) to the detriment of financial Wealth.

However, this sovereign destruction, as skillful as it may be, constitutes only the first half of the topological equation. By drying up the mechanism of real estate pre-sales, the State simultaneously deprived the economy of its historical tool of local monetary creation. Parasitic Wealth is eliminated, but the patient finds himself paralyzed: the economy tips into **Stasis**.

Chapter 2 – The dead-end of the immaterial and temporal avoidance

2.1. Carbon, Living, Cognitive Data: three bubbles without spatial thickness

To avoid social explosion linked to this blockage, the CCP deployed a “Third Mutation” towards the immaterial. The State attempted to simulate a cycle of Extraction by creating normative scarcities (carbon quotas, genomic data exploitation rights, AI tokens). Although these mechanisms faithfully reproduce the mechanics of Extraction (anticipatory credit creation, differential rents), they encounter an ontological obstacle: **the deficit of morphological thickness**.

The algorithm does not generate urban form; it requires neither steel, nor cement, nor mass logistics. Unlike the material frontier, whose ripple effects irrigated the entire economic fabric, the bubbles of Carbon, Living, and Cognitive Data create market value (and commercial surpluses in dollars) without creating territorial use-value for the 1.4 billion Chinese. They are “a-spatial” bubbles that function as palliative shock absorbers, but fail to restart the productive machine.

2.2. The SECC (Computing Capacity Export Strategy): an external palliative facing overproduction

The inability of the immaterial economy to generate domestic demand forced China to export its overcapacity. This is the meaning of the **Computing Capacity Export Strategy (SECC)**.

According to 2026 data, China's installed computing power reached 412 EFLOPS, with a stagnant utilization rate of 62%, revealing a structural overcapacity of 155 EFLOPS. Aware of the West's inability to build its own computing infrastructures (grid connection delays of 5 to 8 years in Europe and the United States), China exports its computing via *Token Outbound*. By publishing *open-source* AI models at derisory inference costs (1 to 30 times lower than Western models), China destroys the incentive for material investment (CAPEX) in the West and captures global demand. Although this SECC generates considerable external rents (gross margins of 72 to 86% for Chinese hyperscalers), it only displaces the problem: revenues accumulate in the balance sheets of tech companies and the State without diffusing into the popular economy.

2.3. The mass employment deficit: elasticity 0.14 against 1.8 for construction

The failure of the Third Mutation is mathematically demonstrable by data from the *China Macroeconomic Forum* (2026). The employment elasticity of the "New Productive Forces" (AI, cloud, robotics sectors) has stabilized at **0.14** (i.e.: 1% growth in digital value added generates only 0.14% growth in employment). In comparison, residential construction had an elasticity of **1.8** during the decade 2010-2020.

The Chinese digital sector today employs only about 21 million people, i.e. **less than a third** of the jobs destroyed by real estate. Worse, the domestic Keynesian multiplier of data centers in Guizhou is only 1.2, compared to 2.5 for construction sites. The immaterial is structurally incapable of absorbing the demand of a population of 1.4 billion. The "New Trio" (AI, robotics, new medicines) confirms China's technological power, but simultaneously validates its social dead-end.

Chapter 3 – The phenomenon of “Tang Ping” as a social symptom

3.1. Withdrawal from competition: sociological data and testimonies

The impossibility of recreating mass employment has translated into a rupture of the implicit social contract of Land Finance, which guaranteed social advancement through effort. This topological blockage manifests socially through the phenomenon of *Tang Ping* (“lying flat”).

This movement is neither a generational whim nor a phenomenon of moral laziness, but a **rational adaptive strategy** facing a blocked system. Sociological data for 2026 are striking: about **45% of young people** declare withdrawing from social competition. Testimonies abound of Tsinghua University graduates accepting security guard positions, or holders of a master's in physics becoming garbage collectors. This behavior reflects the collapse of the return on investment of education. *Tang Ping* manifests as withdrawal from the labor market, forced minimalism of consumption, renunciation of property and starting a family. It is the defection of economic agents minimizing their exposure to losses in a system that no longer rewards them.

3.2. Stasis as an absorbing attractor and risk of long depression

In the topological formalism of the $Cl(6,6)$ algebra, the situation of post-2018 China is modeled as a blockage on an **absorbing attractor**, attractor **S** (dispersion + nodal point). The transition operators (institutional reforms, monetary policy, exogenous shocks) are saturated. The economy can no longer go back (towards polarized growth attractors 3P) because the frontiers are closed, but it also cannot collapse immediately (attractor T) because the Party-State artificially maintains social cohesion through strict algorithmic control and redistribution of rents.

Stasis is this precarious state of equilibrium where the destruction of Wealth is compensated by redistribution, but where the recreation of Value is absent. If Stasis is not treated, it resembles a limit hypothesis that breaks with historical cyclicity. The system becomes mired in a zero-sum game, exposing China to the risk of a long depression (“Japanese-style decline” or “Roman-style decline”).

The composite Stasis index ($\mathcal{F}_{Stase}$), calibrated on the 9 Chinese dynastic declines (now including the periods of fragmentation and Jin/Song split), confirms this diagnosis: China 2026 shows a score of **0.64**, placing it in a state of “characterized Stasis”, on the border of the pre-retropolarity threshold (0.75). Without the activation of a powerful transition operator to cross the topological threshold N_4 , the system exposes itself to a retropolarity (systemic collapse), marking the **10th dynastic decline** of its history.

PART II – THE TOPOLOGICAL ENGINE: GENERAL PRESENTATION

Introduction

Conventional macroeconomics (DSGE models) treats the economy as a continuous space of variables mechanically adjusting towards equilibrium. This approach fails to anticipate abrupt ruptures (1929, 2008, the current Chinese Stasis) because it ignores the discrete and informational nature of complex systems. To steer the exit from the crisis in China, the public decision-maker must have a rigorous cartography of the possible states of the system and its tipping points. The topological model based on the Clifford algebra $Cl(6,6)$ offers this reading grid. It models the economy not as a continuous flow, but as a discrete information system subject to a universal invariant: the reduction $64 \rightarrow 20$.

Chapter 4 – Fundamental principles of modeling

4.1. The economy as a discrete and dynamic information system

The Rent Extraction Theory (TdR) postulates that the economy is not a continuous fluid, but a network of discrete configurations. Just as genetics is based on 64 codons (4 DNA bases combined by 3) which filter information to produce 20 amino acids, the economy obeys strict combinatorial laws. Economic dynamics there are governed by rupture thresholds (the Nodal Points) and by nilpotence — an algebraic principle stipulating that a configuration that violates conservation laws (e.g.: hyperinflation coupled with a collapse in production) is topologically unstable and therefore “forbidden”. This discrete approach makes it possible to diagnose the state of the system in a binary and predictive manner, where continuous models see only fluctuations.

4.2. The universal invariant $64 \rightarrow 20$: applicability to economic regimes

The invariant $64 \rightarrow 20$ is a universal topological law. It manifests in particle physics (64 spin configurations reducing to 20 families), in the genetic code (64 codons for 20 amino acids), in Mandarin phonology, and in the matrimonial alliances of the Tang dynasty. In the economic field, this invariant translates into the fact that 64 theoretical market configurations reduce, by topological filtration (the geometry of the Merkabah coupled with the Clifford algebra), to **20 stable attractors**. These 20 attractors represent the only viable economic regimes in the long term. Any situation outside these 20 attractors is a frustrated configuration, generating systemic instability that will force a return to a stable attractor, often violently (retropolarity).

4.3. The 6 binary variables (Inflation, Purchasing power, Profits, Extraction, Dispersion, Land regime)

The space of 64 configurations is generated by six binary dimensions (0 or 1), forming the backbone of the $Cl(6,0)$ algebra. These variables structure the material reality of the Chinese economy:

1. **Inflation** (0 = asset deflation, 1 = high inflation)
2. **Purchasing power** (0 = stagnant/low, 1 = dynamic/high)
3. **Profits** (0 = falling, 1 = speculative/high)
4. **Extraction** (0 = Nodal point, 1 = active extraction)
5. **Dispersion** (0 = concentration, 1 = fragmentation/ghost cities)
6. **Land regime** (0 = saturation of the ecumene, 1 = spatial expansion)

The extension to the $Cl(6,6)$ algebra introduces 6 “anti-cosmic” variables (of regulation), formalizing the Sheng/Ke dualities (expansion/contraction) of Chinese thought. This generates a Hilbert space of **144 pentads** (12 spectral partitions $\times$ 12 base pentads), offering an unprecedented cartography of the inter-sectoral tensions of the economy.

Chapter 5 – Mapping the 20 dynamic regimes: beyond the Kondratiev cycle

5.1. The pentad triplet as the identity of an economic regime

Unlike the Kondratiev economic cycles which describe long waves empirically and sequentially, the $Cl(6,6)$ topology defines an economic regime by its belonging to a **pentad triplet**. A pentad is an ordered set of 5 algebraic elements coding institutions (Structure = Wood, Earth, Metal), exogenous shocks (Fire), and money (Water). Each stable regime (attractor) is defined by the combination of 3 pentads out of the 12 basic ones. This triadic system makes it possible to model the fine interactions between monetary policy, structural reforms, and external shocks within the same economic configuration, offering a much more precise diagnostic tool than traditional macroeconomic aggregates.

5.2. The polarity gradient $3P \rightarrow 3N$ (from polarized growth to depression)

The 20 attractors are organized according to a **polarity gradient** ranging from 3P (3 positive pentads, pure Sheng mode) to 3N (3 negative pentads, pure Ke mode), passing through mixtures 2P+1N and 1P+2N.

- **3P (Expansive polarized growth):** Rare and unstable regimes of overheating (hyperinflation, massive speculative bubbles).
- **2P+1N (Tempered growth):** The most historically durable regimes, where expansion is regulated.
- **1P+2N (Recession, adjustment, Stasis):** Regimes of tension, majority in times of crisis, where contraction dominates.
- **3N (Terminal depression):** Collapse attractor.

Post-2018 China projects onto attractor **S** (signature 1P+2N: *Dispersion and nodal point*). It is a regime of systemic fragmentation characterized by the triplet $\{P_6, N_5, N_6\}$, reflecting the failure of the consumption recovery (P_6), the exhaustion of resources (N_5), and the blockage of extraction by austerity (N_6).

5.3. The 7 spectral thresholds S_1 to S_7 : the energy gates of the system

The transition from one attractor to another is not continuous. It is constrained by “tropical belts” and crosses **7 critical spectral thresholds** (S_1 to S_7), derived from the eigenvalues of the Λ_{72} lattice. These thresholds are the energy gates of the system:

- S_1 to S_4 : Inflation thresholds, policy tipping points, macroeconomic coordination.
- S_5 (Fire Transition): Activation by exogenous shock / innovation.
- S_6 (Water Transition): Activation by monetary policy / financial reform.
- S_7 (Octave jump): Change of global cycle.

The Topological Golden Rule: Any direct transition between the recession belt (C_N) and the expansion belt (C_P) without passing through the polar thresholds P_4 or N_4 is a violation of nilpotence. It causes a **retropolarity** (systemic collapse). This is the topological equivalent of the “8th decline” (now recalibrated on the historiography of the 9 declines). The Territorial Regeneration Rent (TRR) is precisely the operator that allows this controlled crossing through N_4 .

Chapter 6 – The 4 global observables of Stasis

To pilot this system, the decision-maker has a level 1 (Beijing) spectral dashboard, synthesized by 4 global observables derived from the Dirac operator. These 4 observables are the economic equivalent of the “four examinations” of traditional Chinese medicine (望闻问切 - inspection, auscultation, interrogation, palpation).

6.1. η (asymmetry): the Qi of the system – expansion or recession

The observable η measures the spectral asymmetry between Sheng (expansion) and Ke (regulation/contraction) modes. It is the **Qi** (vital breath) of the economy.

- $\eta > 0$: Energy circulates, the economy is expanding.
- $\eta < 0$: The system is contracting.
- $\eta \approx 0$: Tipping point (pre-bifurcation).

China 2026 Diagnosis: $\eta_{total} \approx -0.45$. The Qi is stagnant, the economy is in marked recession.

6.2. d (spectral dimension): harmony – diversity or concentration

The effective spectral dimension d measures systemic harmony (Shannon entropy). It indicates whether the economy is diversified and resilient, or on the contrary concentrated and rigidified.

- $d \approx 5$: Harmony regime (tempered growth). This is the flourishing of the 5 phases of Wuxing.
- $d < 2$: The system is fragmented, shocks no longer diffuse (red alert).

China 2026 Diagnosis: $d_{total} \approx 2.1$. The economy is excessively concentrated, institutional and territorial diversity is blocked.

6.3. gap (fragility): the margin before collapse

The spectral gap measures the robustness of the system, that is, the margin of maneuver available to the State before hitting a bifurcation threshold. It is the Taoist “creative void” that allows the system to absorb shocks.

- $gap > 0.5$: Robust system.
- $gap < 0.2$: Extreme fragility, close to collapse.

China 2026 Diagnosis: $gap_{total} \approx 0.12$. The system is asphyxiated. The slightest exogenous perturbation (geopolitical, climatic) risks causing an uncontrolled rupture.

6.4. R_{seuil} (tension on thresholds): systemic red alert

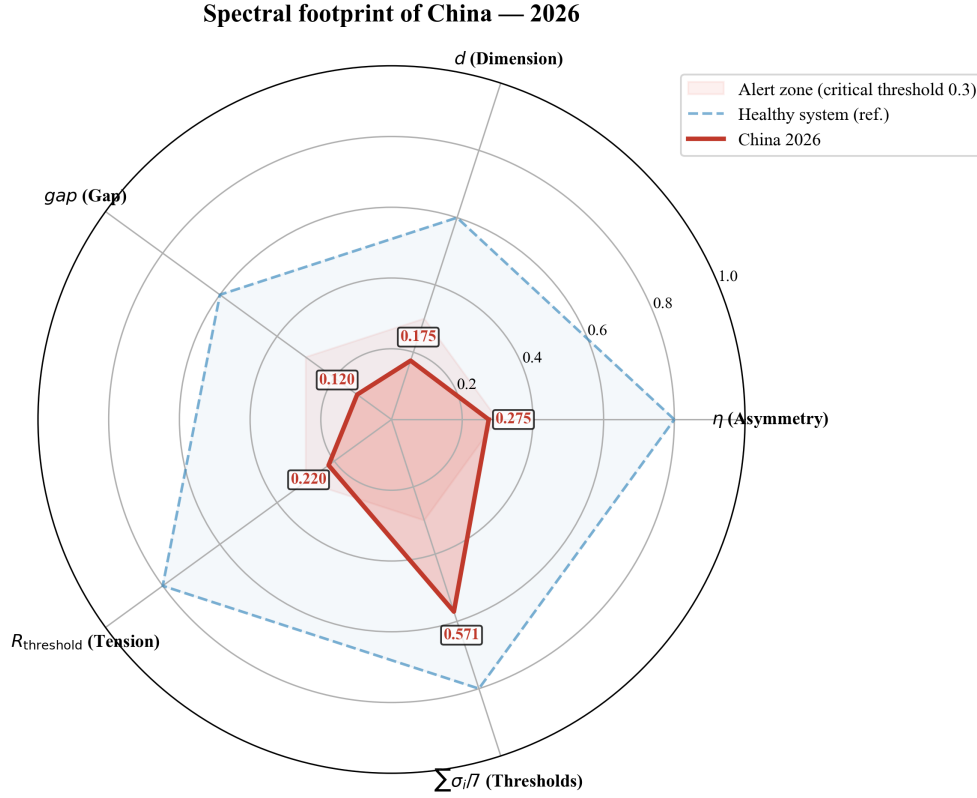
This observable calculates the fraction of spectral energy concentrated on the critical polar thresholds P_4 and N_4 . It indicates whether the system is attempting to force a direct transition from recession to expansion without paying the institutional price (crossing N_4).

- $R_{seuil} < 0.3$: The system is far from the rupture zone.
- $0.3 \leq R_{seuil} \leq 0.7$: Moderate tension.
- $R_{seuil} > 0.7$: **Red alert**, imminent retropolarity (collapse).

China 2026 Diagnosis: $R_{seuil} \approx 0.78$. The system is in pre-retropolarity. The transition operators are saturated. The window for political action to avoid the 10th decline is extremely narrow.

6.5. Visual Synthesis: The Spectral Footprint

The 4 vital signs (η , d , gap , $R_{\text{threshold}}$) and the aggregated threshold $\sum \sigma_i/7$ can be visualized on a **spectral radar**, providing an immediate reading of the system's state.



Spectral footprint of China —2026. The red polygon represents the current state of the system; the dashed blue polygon represents a healthy reference system; the regular pentagon at 0.3 (light red) delimits the alert zone below which each dimension is in critical crisis. Four of the five dimensions (η , d , gap , $R_{\text{threshold}}$) are below this threshold, confirming the diagnosis of characterized Stasis.

Reading the radar:

- η (**Asymmetry**): 0.275 → the Qi is in moderate contraction (far from the 0.7 of a healthy system).
- d (**Dimension**): 0.175 → severe fragmentation (a healthy system would be around 0.6).
- gap (**Margin**): 0.120 → extreme fragility (a healthy system is around 0.5).
- $R_{\text{threshold}}$ (**Tension**): 0.220 → moderate tension on the hinges (but a healthy system would be around 0.2).
- $\sum \sigma_i/7$ (**Thresholds**): 0.571 → more than half of the spectral thresholds have been crossed.

Synthetic diagnosis: The red polygon is **flattened** toward the center, meaning the system is **locked** on a Stasis attractor. The contrast with the blue polygon (healthy system) is maximal on the d and gap axes.

Summary of Part II: The topological diagnosis confirms Stasis. The system is locked on attractor S , with extreme fragility and maximum tension on thresholds. Mathematical modeling proves that a classic monetary stimulus or a direct technological jump towards the immaterial would violate the threshold rule and cause a collapse. Exiting Stasis imperatively requires the controlled activation of a transition operator capable of crossing the N_4 threshold. This operator is the TRR, which will be the subject of Part IV.

PART III – THE ADMINISTRATIVE ARCHITECTURE OF THE MODEL

Introduction

The relevance of a topological model for a strategic think tank does not lie in its mathematical purity, but in its capacity to be operationalized by the State. The Cl(6,6) model does not merely diagnose Stasis; it provides a “nested” steering grid — that is, deeply integrated into the administrative organization of China. It translates the physics of complex systems into a four-level decision-making dashboard, from the Politburo down to county magistrates, via provincial governors.

Chapter 7 – A steering grid nested within the State organization

7.1. Resonance principle: granularity of diagnosis = granularity of intervention

The major innovation of this steering architecture is the **resonance principle**. In traditional Chinese medicine (TCM), one does not apply the same acupuncture needle to treat a general imbalance of Qi or a localized blockage of a specific meridian. Likewise, economic regulation requires that the scale of political intervention correspond to the granularity of the spectral diagnosis. If the State Council (Beijing) examines macroeconomic aggregates, it is blind to sectoral dissonances. Conversely, if a county magistrate adjusts local policies without a systemic vision, he risks aggravating fragmentation. The model therefore deploys a dashboard with 5 levels of observables, strictly articulated with the 4 decision-making layers of the Party-State.

7.2. The 12 spectral leaves (meridians) and their correspondence with economic sectors

The extension to the Cl(6,6) algebra decomposes the system into 12 “spectral leaves”, which are the economic equivalents of the 12 meridians of TCM. Each leaf isolates the specific dynamics of a sector or a key variable, allowing the diagnosis of internal tensions:

- **Cosmic leaves (Sheng mode, expansion $\eta > 0$):**
 - e_1 : Innovation (R&D, technologies)
 - e_2 : Purchasing power (Domestic consumption)
 - e_3 : Profits (Productive accumulation)

- e_4 : Active extraction (Value transfer, securitization)
 - e_5 : Dispersion (Frontier, migratory flows)
 - e_6 : Land (Territorial regime)
 - **Anti-cosmic leaves (Ke mode, regulation $\eta < 0$):**
 - f_1 : Wage regulation
 - f_2 : Financial regulation (Credit, exchange rates)
 - f_3 : Profit regulation (Taxation, LGFV rents)
 - f_4 : Extraction regulation
 - f_5 : Dispersion regulation
 - f_6 : Land regulation (Real estate vacancy)
-

Chapter 8 – Level 1: Strategic steering (Beijing)

8.1. The 4 global observables for the State Council

The strategic layer (State Council, People’s Bank of China, NDRC) uses the Level 1 dashboard, aggregating the 12 leaves into 4 macroscopic observables (η_{total} , d_{total} , gap_{total} , $R_{seuil,total}$). This level provides the global “photograph” of the system’s health. Beijing does not need to know which specific pentad is saturating in Xingguo county; it needs to know whether the Mandate of Heaven is globally preserved.

8.2. Decision rules: when to activate the TRR, when to reform the currency

The Level 1 observables trigger strict strategic decision rules:

- If $\eta < -0.4$ AND $d < 2$: The system is in characterized Stasis and fragmented. Rule: **Monetary reform** (massive securitization of GEP to circumvent LGFV debt).
- If $gap < 0.1$: Extreme fragility. Rule: **Emergency rescue plan** (monetization of ecological Value).
- If $R_{seuil} > 0.7$ AND $gap < 0.2$: Red alert, imminent retropolarity. Rule: **National activation of the TRR**.

8.3. The composite Stasis index $\mathcal{F}_{Stase}$: the quantified “Mandate of Heaven”

To synthesize these signals, the model constructs a composite Stasis index ($\mathcal{F}_{Stase}$), which quantifies the “Mandate of Heaven”. It combines the 5 levels of observables according to weights (w_i) calibrated on the historiography of the **9 dynastic declines** (including the fragmentations of the Five Barbarians and Jin/Song).

The decision thresholds for Beijing are:

- $\mathcal{F}_{Stase} < 0.3$: Healthy growth (Green).

- $0.3 \leq \mathcal{F}_{Stase} < 0.65$: Moderate tension (Orange).
- $\mathcal{F}_{Stase} \geq 0.65$: **Characterized Stasis (Red)**. Mandatory activation of the TRR.

China 2026 Diagnosis: $\mathcal{F}_{Stase} = 0.644$. The country is on red alert. The activation of the Territorial Regeneration Rent is no longer an ecological option, it is a necessity of political survival.

Chapter 9 – Level 2: Tactical steering (Provinces)

9.1. The 48 observables per spectral leaf ($\eta_f, d_f, gap_f, R_{seuil,f}$)

The tactical layer (Provincial Governors, regional NDRC offices) uses Level 2: the 12 spectral leaves, each with its 4 local observables (i.e., 48 indicators). This level reveals **contextual dissonances**. A province may show an acceptable η_{total} , while hiding a localized collapse on a critical leaf (e.g.: f_3 LGFV Accumulation).

9.2. Adjustment of GEPs (Gross Ecosystem Product) and CURs by region

Governors adjust sectoral levers according to pathological leaves:

- If leaf e_3 (Profits) is in strong Ke ($\eta_{e_3} < -0.5$): Local fiscal reform, taxation of speculative rents, redistribution towards GEP.
- If leaf f_3 (Debt accumulation) saturates ($R_{seuil,f_3} > 0.7$): Restructuring of local LGFVs, issuance of Regenerative Usufruct Certificates (CUR) backed by regional ecological assets.

9.3. Case of Zhejiang: imbalance between Innovation (e_1) and Accumulation (f_3)

The province of Zhejiang perfectly illustrates the usefulness of Level 2. In 2026, the Innovation leaf (e_1) shows moderate expansion ($\eta_{e_1} = +0.3$), suggesting that the technology sector (Alibaba, etc.) is doing well. However, the Accumulation leaf (f_3) is in deep contraction ($R_{seuil,f_3} = 0.82$), reflecting the asphyxiation of local land platforms.

* **Tactical action:** The governor does not relaunch innovation (which creates few jobs), but pilots the Two Mountains Bank in Lishui to securitize the forest GEP. This recapitalizes provincial LGFVs by backing the debt to a real asset (ecological regeneration), thus rebalancing f_3 without killing e_1 .

Chapter 10 – Level 3: Operational steering (Counties / Pilot zones)

10.1. The 144 projected pentads: the economy in “acupuncture points”

At the county level, macroeconomic management becomes “economic acupuncture”. The system projects itself onto the **144 pentads** (12 base pentads $\times$ 12 spectral partitions). The county magistrate identifies the specific pentad that constitutes the node of the local blockage.

10.2. Activity, polarity, connectivity, frustration – the 4 signs of the pentad

Each pentad is evaluated by 4 ultra-fine local observables:

- **Activity** ($a_{k,g}$): Intensity of the configuration.
- **Polarity** ($\pi_{k,g}$): Sheng mode (+1) or Ke (-1).
- **Connectivity** ($c_{k,g}$): Number of links with other configurations.
- **Frustration energy** ($E_{k,g}$): Deviation from the sectoral average.

A pentad whose activity is > 0.7 , connectivity strong, and frustration low, is the **critical node**. Desaturating this node restores the circulation of local economic Qi.

10.3. Case of Lishui: desaturation of node $N_5^{(f_3)}$ by the Two Mountains Bank

In the municipality of Lishui, the Level 4 diagnosis identified pentad $N_5^{(f_3)}$ (Stasis / Regulated Accumulation) as the critical node ($a = 0.78$, $c = 5$). This pentad reflected the blockage of impoverished agricultural lands and abandoned villages.

- **Operational action:** The magistrate deployed the Two Mountains Bank. This institution aggregated the use rights of these degraded lands, issuing loans backed by these assets to finance ecological restoration. The credit injected created a massive “advanced exogenous demand” in the form of local rural jobs.
- **Spectral result:** The activity of pentad $N_5^{(f_3)}$ was desaturated (targeting a drop from 0.78 to 0.40), unblocking the circulation of value towards the expansion belt (C_P) without resorting to concretization.

Chapter 11 – Macro-systemic governance: the National Observatory dashboard

11.1. The 7 spectral thresholds as an early warning system

The transversal layer (National Spectral Monitoring Center, research academies) uses Level 3: the 7 spectral thresholds (S_1 to S_7). This level acts as an Early Warning System. The threshold vector $\vec{\sigma}$ codes the crossing state of the system. China 2026 shows $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$. The first 4 thresholds (exhaustion of the previous model) are crossed, but the system is **blocked between S_4 and S_5** . The National Observatory must alert Beijing if the topological tension $\mathcal{T}$ approaches the threshold $S_5 = 0.17195$ (Fire Transition, ecological innovation), the *sine qua non* condition to initiate the TRR.

11.2. Wuxing tensions: rebalancing the 5 phases (Wood, Fire, Earth, Metal, Water)

The Observatory also groups the 12 leaves into 5 families corresponding to Wuxing (the Five Phases), to diagnose structural imbalances:

- Wood (Innovation) / Fire (Purchasing power) / Earth (Profits) / Metal (Extraction) / Water (Land).

China 2026 Diagnosis:

- $\eta_{\text{Metal}} = -0.70$ (Extraction blocked) smothers $\eta_{\text{Wood}} = -0.05$ (Innovation).
- The maximum gap $\Delta\eta_{\text{max}} = 1.0$ between Fire and Metal signals extreme tension.
- The generative (Sheng) cycle is broken: Earth (Profits) no longer nourishes Metal.
- The Observatory prescribes the TRR to restore the cycle: *Wood (ecological innovation) → Fire (green jobs) → Earth (territorial GEP) → Metal (CUR) → Water (basin regeneration)*.

11.3. The inter-leaf correlation matrix: detecting regional fragmentation

Finally, the Observatory calculates the correlation matrix $C(t)$ between the 12 leaves, measuring whether the country is coherent or bipolar.

China 2026 Diagnosis:

- The matrix reveals a divided system. A Sheng block (Innovation, Purchasing power, strongly correlated at +0.70) evolves in total opposition to a Ke block (Profits, Accumulation, Land, correlated among themselves but negatively with the Sheng block).
 - The fragmentation index $\mathcal{F} = 0.42$ confirms this cleavage.
 - The lesson for the decision-maker is clear: China is not simply in recession, it is torn between two economies that no longer communicate (the data economy and the material economy).
 - Only the TRR, by re-anchoring cognitive value in regenerated material space, can restore this matrix.
-

PART IV – THE TRR AS A TRANSITION OPERATOR

Introduction

Chinese Stasis is not a historical fatality, but a precise topological state: the system is locked on attractor S (dispersion + nodal point). Classic stimulus tools (monetary policies, industrial subsidies) are saturated and any attempt to “jump” directly to the expansion phase (C_p) without passing through the polar thresholds would trigger retropolarity (systemic collapse). The only topologically authorized trajectory to exit the crisis is the controlled crossing of the threshold N_4 . The operator capable of accomplishing this passage is the **Territorial Regeneration Rent (TRR)**.

Chapter 12 – Formalization of the Territorial Regeneration Rent (TRR)

12.1. The TRR as a controlled crossing through the threshold N_4

In the Cl(6,6) model, the TRR is not a simple ancillary ecological policy. It is the topological operator T_{mixed} that allows the system to leave the recession belt (C_N) and join the expansion belt (C_P) via a new material Extraction cycle. Instead of exhausting new virgin lands (which is impossible, the spatial Nodal Point having been reached), the TRR creates Value by regenerating the space degraded by forty years of hyper-industrialization. It substitutes the destruction of Wealth by an endogenous creation of Value.

12.2. The operator T_{mixed} : combining $T_{\text{structure}}$, T_{fire} , T_{water}

The activation of the threshold N_4 requires a simultaneous systemic intervention on the three components of the economic pentad, formalized by the mixed transition operator (T_{mixed}):

- $T_{\text{structure}}$ (**Institutions**): Creation of a new productive model based on territorial ecology.
- T_{water} (**Money/Credit**): Reorientation of financial flows, circumventing the LGFV debt impasse.
- T_{fire} (**Shocks/Innovation**): Use of the climate crisis as a catalyst to force the transition.

12.3. The 3 structural reforms as components of the engine

For the public decision-maker, the operator T_{mixed} translates into three inseparable structural reforms, each targeting a specific spectral threshold:

1. **The Two Mountains Bank (Component T_{water})**: This monetary mechanism securitizes the Gross Ecosystem Product (GEP). It allows circumventing the asphyxiating debt of local governments (LGFVs) by backing new credit to restored ecological assets. It allows crossing the threshold S_6 (Water Transition).
2. **Integration of GEP into local KPIs (Component $T_{\text{structure}}$)**: A major institutional reform, the replacement of GDP by GEP as a performance indicator for local cadres anchors the new productive model in the bureaucracy. It allows crossing the threshold S_5 (Fire Transition).
3. **The Regenerative Usufruct Certificate - CUR (Component T_{fire})**: This financial instrument returns the appropriation of the rent to local communities on the lands they have regenerated. It creates a new value distribution cycle and triggers the global cycle change (crossing threshold S_7 , the octave jump).

Chapter 13 – Pilot zones: the TRR already at work

Unlike many macroeconomic theories, the TRR is not a laboratory hypothesis. The Party-State has already begun to deploy it empirically on the ground, through programs officially called “Mechanisms for realizing the value of ecological products” or “Two Mountains Banks”.

13.1. Lishui (Zhejiang): Two Mountains Bank and GEP securitization

Lishui is the integrated epicenter of the model. The municipality created a platform where the Two Mountains Bank aggregates the use rights of impoverished agricultural lands and abandoned villages. It issues loans backed by these assets. The injected credit finances ecological restoration, generating a massive “advanced exogenous demand” in the form of local rural jobs. It is the perfect incarnation of the new material Extraction.

13.2. Sanming / Nanping (Fujian): Lin-piao and regenerated forests

In these cities, the problem concerned fragmented and non-marketable collective forests. State-owned enterprises buy the management rights, inject anticipatory credit to hire locals to replant and sustainably exploit the forest. In exchange, villagers receive “Lin-piao” (forestry tickets), negotiable financial instruments, creating a green differential rent.

13.3. Xingguo / Ganzhou (Jiangxi): watersheds and contaminated soils

This area faces eroded watersheds and soils contaminated by rare earth mining. The water TRR mechanism is deployed there: upstream cities receive state funding and “ecosystem service payments” from downstream cities (which benefit from clean water) to decontaminate and restore the soils.

13.4. Hainan: island GEP and prohibition of speculation

At the provincial level, the island of Hainan has purely and simply banned speculative real estate construction on its coastline. The GEP is used as the exclusive indicator to evaluate local officials and as collateral to obtain massive bank loans, transforming the entire island into a regeneration frontier.

Chapter 14 – Deployment schedule 2026-2035

The operational schedule of the TRR, aligned with the Five-Year Plans, corresponds to a rigorous sequence of crossing spectral thresholds (S_5, S_6, S_7).

14.1. Pilot phase (2024-2025): crossing S_5 (Fire)

- **Objective:** Test mechanisms on a small scale (10 pilot regions in 2024, 25 in 2025).
- **Action:** Launch of the Two Mountains Bank in Lishui, integration of GEP into local KPIs.
- **Spectral target:** Crossing threshold S_5 (Fire Transition). The observable η must go from -0.45 to -0.35; R_{seuil} from 0.75 to 0.65.

14.2. Provincial deployment (2026-2028): crossing S_6 (Water)

- **Objective:** Extend the model to validated provinces.
- **Action:** Generalization of CURs and GEP securitization. Treatment of 2 to 5 million hectares.
- **Spectral target:** Crossing threshold S_6 (Water Transition). The observable η goes from -0.35 to -0.15; R_{seuil} from 0.65 to 0.45. Reform of the monetary system.

14.3. Acceleration (2029-2030): crossing S_7 (Octave jump)

- **Objective:** Scale up nationally.
- **Action:** Massive mobilization of capital, treatment of 5 to 10 million cumulative hectares.
- **Spectral target:** Crossing threshold S_7 (Octave jump). The observable η becomes positive (+0.10); R_{seuil} drops to 0.35. Exit from Stasis.

14.4. Generalization (2031-2035): stabilization on C_P (new Extraction cycle)

- **Objective:** Full coverage of atrophied rural spaces.
- **Action:** Consolidation of sectors (agrivoltaics, aquaculture, reforestation).
- **Spectral target:** Stabilization on the expansion belt (C_P). The Stasis index $\mathcal{F}_{Stase}$ must fall below the threshold of 0.3, marking a return to healthy growth.

Chapter 15 – Simulation of the two scenarios

For the strategic decision-maker, the Cl(6,6) model makes it possible to simulate the evolution of spectral observables over the period 2026-2032, confronting inaction (or the immaterial flight forward) with the strict deployment of the TRR.

15.1. Scenario 1: Without TRR – retropolarity and tenth decline (observables 2032)

If the system attempts a direct transition towards expansion without passing through the threshold N_4 (for example, by a classic monetary stimulus or an injection of liquidity into real estate), it violates nilpotence.

- **2028** ($\mathcal{F}_{Stase} = 0.70$): Pre-retropolarity.
- **2030** ($\mathcal{F}_{Stase} = 0.75$): Imminent collapse.
- **2032** ($\mathcal{F}_{Stase} = 0.80$): **Retropolarity (10th dynastic decline).**
- **Observables 2032:** $\eta \rightarrow -0.80$ (total collapse of Qi), $d \rightarrow 0.8$ (extreme fragmentation), $gap \rightarrow 0.02$ (absolute fragility). The threshold vector $\vec{\sigma}$ becomes (1, 1, 1, 1, 1, 1): the octave jump is undergone destructively. This is the repetition of the fatal cycle of the 9 historical declines.

15.2. Scenario 2: With TRR – exit from Stasis and healthy growth (observables 2032)

If the TRR is activated according to the schedule (sequential crossing of S_5, S_6, S_7):

- **2028** ($\mathcal{F}_{Stase} = 0.48$): Emerging Stasis, beginning of recovery.
- **2030** ($\mathcal{F}_{Stase} = 0.32$): Moderate tension, exit from Stasis.
- **2032** ($\mathcal{F}_{Stase} = 0.22$): **Healthy growth.**
- **Observables 2032:** $\eta \rightarrow +0.35$ (Qi circulates abundantly), $d \rightarrow 4.2$ (harmony and diversity restored), $gap \rightarrow 0.45$ (system robustness). Wuxing is balanced, the Sheng (generative) cycle is restored.

15.3. Synthetic comparison of indicators

The 2032 dashboard illustrates the radical asymmetry of the two trajectories:

Table 3: 2032 Dashboard –Comparison of trajectories

Indicator (2032)	Scenario 1 (Without TRR)	Scenario 2 (With TRR)	Stabilization Delta
η_{total} (Direction)	-0.80	+0.35	+1.15
d_{total} (Harmony)	0.8	4.2	+3.4
gap (Robustness)	0.02	0.45	+0.43
R_{seuil} (Tension)	0.95	0.25	-0.70
$\mathcal{F}_{Stase}$ (Global index)	0.80 (Retropolarity)	0.22 (Healthy growth)	-0.58

Conclusion: The simulation demonstrates that the TRR is not an option among others, but the only topological operator allowing the reversal of the trend. Without it, the Chinese economy inevitably tips towards a retropolarity (the 10th decline) by 2032. With it, China returns to a regime of healthy growth, comparable to the Thirty Glorious Years, this time based on the material regeneration of its own territory.

PART V – STRATEGIC DASHBOARD

Introduction

The operational contribution of the Cl(6,6) model for the public decision-maker materializes in a multi-level Strategic Dashboard. It transforms the physics of complex systems into a daily governance tool. Far from being a mathematical black box, this dashboard is the modern equivalent of the “four examinations” (望闻问切) of traditional Chinese medicine: it allows taking the pulse of the economic system at five levels of resolution, from the global view (Beijing) to precise acupuncture points (Counties). This chapter restores the spectral photograph of China in August 2026, on the eve of the effective launch of the 15th Five-Year Plan (2026-2030).

Chapter 16 – The multi-level dashboard (China 2026)

The spectral diagnosis of post-2018 China deploys over five nested levels. The global table synthesizes the systemic red alert and the priority intervention points.

Multi-level spectral dashboard —China August 2026

LEVEL 1 —GLOBAL VIEW (4 observables integrated over 12 leaves)

η_{total}	d_{total}	gap	R_{seuil}	$\mathcal{F}_{\text{Stase}}$
-0.45	2.1	0.12	0.78	0.64

State: **RED ALERT** —Characterized Stasis

LEVEL 2 —DOMINANT LEAVES (48 observables per leaf)

Leaf	η	d	gap	R_{seuil}
e_3 (Profits)	-0.70	1.8	0.08	0.82
f_3 (Accumulation)	-0.60	2.0	0.10	0.75
f_6 (Land)	-0.55	1.7	0.07	0.80
e_1 (Innovation)	+0.30	3.5	0.25	0.40
e_2 (Purchasing power)	+0.25	2.8	0.18	0.45

LEVEL 3 —ACTIVATED THRESHOLDS (7 spectral thresholds from Λ_{72})

Threshold vector $\vec{\sigma}$	Crossing state
(1, 1, 1, 1, 0, 0, 0)	S_1, S_2, S_3, S_4 crossed (exhaustion of previous model) S_5, S_6, S_7 blocked (no new cycle)

Topological tension $\mathcal{T} = 0.165$ | Next threshold $S_5 = 0.17195$ (*not reached*, $\Delta = 0.007$)

LEVEL 4 —ACTIVE PENTADS (144 projected pentads)

Pentad	Activity	Connectivity	Frustration	Role
$N_5^{(f_3)}$	0.78	5	0.08	Critical node
$N_6^{(f_3)}$	0.65	4	0.15	Contagious
$P_6^{(e_2)}$	0.45	3	0.33	Frustrated
$P_1^{(e_1)}$	0.35	2	0.20	Emerging

LEVEL 5 —WUXING TENSIONS (66 relational observables)

	η_{Wood}	η_{Fire}	η_{Earth}	η_{Metal}	η_{Water}
Value	-0.05	-0.15	$_{26}$ -0.55	-0.70	-0.52

Structural imbalance: Metal (Extraction) smothers Wood (Innovation).

$\Delta\eta_{\text{max}} = 1.0$ (between Fire and Earth) | Fragmentation $\mathcal{F} = 0.42$ (bipolar Sheng/Ke system)

16.1. Level 1 – Beijing Radar

The Beijing Radar offers the global view. The four integrated observables confirm the diagnosis made in Part I: China is in characterized Stasis on attractor S. The composite index $\mathcal{F}_{\text{Stase}} = 0.64$ places the system at the border of pre-retropolarity (critical threshold 0.75). This level validates the impossibility of a recovery through classic monetary tools and imposes the immediate activation of the TRR.

The radar footprint of China in 2026 is characteristic of a system in pre-retropolarity: the margin of maneuver (gap) and asymmetry (η) are collapsed, while the tension on thresholds (R_{seuil}) has reached a critical level at 0.78. The spectral dimension (d) collapses, reflecting the fragmentation of the system. Compared to the reference footprint of a healthy system (Thirty Glorious Years), the imbalance is total, justifying the immediate activation of the emergency protocol (TRR).

16.2. Level 2 – Heat map of the 12 spectral leaves

Level 2 reveals the **bipolar signature** of Chinese Stasis. The material leaves (e_3 Profits, f_3 Accumulation, f_6 Land) are in deep Ke contraction ($\eta < -0.5$), reflecting the asphyxiation of land finance. Conversely, the cognitive leaves (e_1 Innovation, e_2 Purchasing power) attempt moderate Sheng expansion. This imbalance is the spectral translation of the spatial thickness deficit: the cognitive rent from the Computing Capacity Export Strategy (SECC) does not diffuse towards material wages, widening the system's fracture.

The heat map below materializes this local asymmetry (η_f) for the 12 meridians of the Chinese economy in 2026:

Table 4: Heat map of the 12 spectral leaves —China August 2026

Cosmic Leaves (e)	e_1	e_2	e_3	e_4	e_5	e_6
Sector	Innovation	Purchasing power	Profits	Active extraction	Dispersion	Land
Asymmetry (η_f)	+0.30	+0.25	-0.70	-0.80	-0.50	-0.55
Anti-cosmic Leaves (f)	f_1	f_2	f_3	f_4	f_5	f_6
Sector	Wage reg.	Financial reg.	Profit reg.	Extraction reg.	Dispersion reg.	Land reg.
Asymmetry (η_f)	-0.40	-0.60	-0.80	-0.60	-0.40	-0.50

Reading: **Orange** (Sheng moderate, $\eta > 0$) / **Light blue** (Ke moderate, $-0.6 \leq \eta \leq -0.3$) / **Dark blue** (Ke strong, $\eta < -0.6$).

The maximum gap ($\Delta\eta_{\max} = 1.10$) between leaf e_1 (Innovation) and leaf e_4 (Active extraction) confirms the total disconnection between the technology sector and the real engine of the economy.

16.3. Level 3 – Threshold vector

The threshold vector $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$ indicates that China has crossed the first four spectral thresholds (exhaustion of the previous growth model) but remains **blocked between S_4 and S_5** . The topological tension ($\mathcal{T} = 0.165$) is insufficient to cross the threshold S_5 (Fire Transition). This blockage signals the absence of a

creative exogenous shock. The TRR precisely has the function of providing this trigger to cross S_5 , S_6 , and then S_7 .

16.4. Level 4 – Active pentads

This level identifies the economic “acupuncture points”. Pentad $N_5^{(f_3)}$ (Stasis with strict profit regulation) is the **critical node**: its activity is maximal (0.78), its connectivity strong (5), and its frustration minimal (0.08). It concentrates all the blockage dynamics of Stasis. The topological prescription is clear: any political intervention (CUR) must prioritize the desaturation of this node N_5 .

16.5. Level 5 – Wuxing tensions

The analysis of Wuxing tensions confirms the cultural and structural diagnosis. The Ke (regulatory) cycle dominates: Metal ($\eta = -0.70$, extraction blockage) smothers Wood ($\eta = -0.05$, innovation). Earth (profits) no longer nourishes Metal. The maximum gap of 1.0 between Fire and Metal signals extreme tension between the cognitive periphery and the central financial regulation.

Chapter 17 – Traffic light system and emergency procedures

The aggregation of multi-level observables produces a hierarchical traffic light system, intended to trigger specific intervention protocols at each layer of the State.

17.1. Alert thresholds by administrative level

Table 5: Hierarchical traffic light system (multi-level aggregation)

Level (Decision-maker)	Green	Orange	Red
1. Strategic (Beijing)	$\mathcal{F}_{\text{Stase}} < 0.3$	$0.3 \leq \mathcal{F}_{\text{Stase}} < 0.65$	$\mathcal{F}_{\text{Stase}} \geq 0.65$
2. Tactical (Province)	$\eta_f > +0.3$	$-0.3 \leq \eta_f \leq +0.3$	$\eta_f < -0.3$
3. Operational (County)	$a_{k,g} < 0.5$	$0.5 \leq a_{k,g} < 0.7$	$a_{k,g} \geq 0.7$
4. Diagnostic (Center)	$\Delta\eta_{\max} < 0.5$	$0.5 \leq \Delta\eta_{\max} < 1.0$	$\Delta\eta_{\max} \geq 1.0$

Aggregation rule: The global alert level is the maximum of the alert levels of the 4 layers. In 2026, China is in the red on levels 1, 2, 3, and 4, triggering the Red Zone Intervention Protocol.

17.2. Intervention protocol in the red zone

When the red alert is triggered (system blocked in $S < 0$ and $R_{\text{seuil}} > 0.7$), the decision-maker must no longer attempt to adjust continuous macroeconomic variables (rate cuts, QE), which would violate nilpotence. He must activate the emergency protocol provided by the Cl(6,6) model:

1. **Verification:** Consult the multi-level dashboard to confirm the saturation of threshold N_4 .

2. **Analysis:** Identify the current attractor (S) and the critical node (N_5).
3. **Decision:** National launch of the operator T_{mixed} (Activation of the TRR).
 - *Strategic Layer:* Decree integrating the GEP into national KPIs.
 - *Tactical Layer:* Opening of credit lines of the Two Mountains Bank at the provincial level.
 - *Operational Layer:* Issuance of Regenerative Usufruct Certificates (CUR) targeting overexpressed pentads.
4. **Monitoring:** Surveillance of the rise of the observable η (target: $\eta > -0.15$ by 2028).

17.3. Monitoring indicators for the 15th Five-Year Plan (2026-2030)

For the 15th Five-Year Plan, traditional steering indicators (GDP, inflation) are enriched by spectral observables. The targets for 2030 for the State Council are as follows:

Table 6: Spectral steering indicators for the 15th Five-Year Plan (2026-2030)

Observable	Value 2026	Target 2028	Target 2030	Threshold to cross
η_{total} (Direction)	-0.45	-0.15	+0.10	S_6 (Water)
d_{total} (Harmony)	2.1	2.8	3.5	Desaturation N_5
gap (Robustness)	0.12	0.25	0.35	Reopening of the network
R_{seuil} (Tension)	0.78	0.45	0.35	Crossing S_7
$\mathcal{F}_{\text{Stase}}$ (Global index)	0.64	0.48	0.32	Exit from Stasis

This dashboard provides the Party-State with an unprecedented governance tool: it quantifies the restoration of the Mandate of Heaven not by GDP growth, but by the restoration of the circulation of economic Qi (η), structural harmony (d), and the distancing from the collapse zone (gap and R_{seuil}).

PART VI – STRATEGIC AND GEOPOLITICAL IMPLICATIONS

Introduction

The Territorial Regeneration Rent (TRR) is not a simple sectoral ecological transition policy. Strategically, it is the regime's survival operator. It responds to a triple requirement: preserve internal political legitimacy (the Mandate of Heaven), inscribe itself in the cultural continuity of Chinese civilization to ensure its acceptability, and redefine China's geopolitical posture by ending the impasse of spatial expansion. The Cl(6,6) model reveals that the exit from Stasis is not only a macroeconomic necessity, but an absolute condition for the stability of the Chinese world-system.

Chapter 18 – The TRR and the preservation of the Mandate of Heaven

18.1. The Mandate as an implicit social contract

Within the framework of the morphogenetic trajectory, the Mandate of Heaven (*Tianming*) is not a cosmological abstraction. It is an **implicit social contract** between the Party-State and the population: the State ensures prosperity, stability, and social harmony; the population grants its legitimacy to the State. If the State no longer fulfills its part (falling prosperity, mass unemployment, growing inequalities), the Mandate erodes and coercion costs increase. The alternative paths of polarized growth 2.0 or neo-feudal retreat threaten this contract: the first destroys mass employment (triggering *Tang Ping*), the second fractures the territory. The Mandate of Heaven is today quantified by the Stasis index ($\mathcal{F}_{Stase}$); an index above 0.7 signals an imminent loss of the Mandate (retropolarity).

18.2. The TRR as a restoration of political virtue (德, De)

The TRR is the only path that concretely repairs the damage of the previous accumulation cycle: ghost cities are converted, atrophied countryside regenerated, water tables restored. In doing so, it restores the **political virtue** (*De*) of the sovereign — the legitimacy based on the capacity to ensure harmony between Heaven, Earth, and Men. The conservation law of value is respected no longer by violent destruction (revolts, wars), but by endogenous value creation.

18.3. Response to the demographic crisis: mass employment and living territories

The falling birth rate and aging reduce the available workforce. The TRR responds to this crisis through territorial anchoring:

1. **Rural mass employment:** young people find productive work in regenerated countryside (agrivoltaics, aquaculture, ecotourism) without migrating to saturated metropolises.
2. **Attractiveness of territories:** vibrant countryside, equipped with services and schools, encourages birth rates.
3. **Reduction of reproduction costs:** ecological regeneration (water, soils, energy) lowers the cost of living, relieving pressure on households.

Chapter 19 – Cultural correspondences as a framework for legitimation

19.1. Yi Jing: 64 hexagrams, 64 configurations – the modernized divinatory dialogue

The epistemological innovation of the Cl(6,6) model is not a deracinated Western import, but the **mathematical formalization of correlative Chinese thought**. The Yi Jing is based on 64 hexagrams (6 binary traits), strictly isomorphic to the 64 economic configurations. The divinatory practice (drawing sticks) aimed to probe the state of the system to adjust policies. The modern spectral dashboard (η, d, gap, R_{seuil}) is the operational equivalent of this state divination: it takes the system's pulse in real time.

19.2. Wuxing: the 5 phases as regulation of Sheng/Ke cycles

The theory of the Five Phases (Wood, Fire, Earth, Metal, Water) describes the regulation of Sheng (generative/-expansion) and Ke (regulatory/contraction) cycles. The Cl(6,6) model formalizes these cycles by Sheng modes ($\eta > 0$) and Ke modes ($\eta < 0$). Chinese Stasis is topologically a Wuxing imbalance where Metal (Extraction) smothers Wood (Innovation). The TRR restores the Sheng cycle: *Wood (ecological innovation) → Fire (green jobs) → Earth (GEP) → Metal (CUR) → Water (regeneration)*.

19.3. Traditional Chinese Medicine: 12 meridians, 4 examinations

TCM postulates that illness is a blockage of the circulation of *Qi* in the 12 meridians. The isomorphism with the 12 spectral leaves is perfect. The Level 2 strategic dashboard reproduces the “four examinations” of TCM (望闻问切):

- η (**Inspection**): the local *Qi*.
- $\Delta\eta$ (**Auscultation**): the Wuxing imbalance.
- $C_{a,b}$ (**Interrogation**): the crossed Sheng/Ke cycles.
- **Active pentads (Palpation)**: the acupuncture points to stimulate.

Table 7: Synthesis of correspondences between Chinese structural wisdom and the Cl(6,6) formalism

Chinese Tradition	Key Concept	Cl(6,6) Correspondence	Operational Dimension
<i>Yi Jing</i>	64 hexagrams	64 economic configurations	Space of possibilities
<i>Yi Jing</i>	Mutations	Transitions between attractors	Dynamics of cycles
<i>Yi Jing</i>	Divination	Spectral observables	Real-time diagnosis
<i>Wuxing</i>	5 phases	Pentad $\{B_1, B_2, B_3, F, S\}$	Elementary structure
<i>Wuxing</i>	Sheng cycle	Sheng mode ($\eta > 0$)	Expansion
<i>Wuxing</i>	Ke cycle	Ke mode ($\eta < 0$)	Regulation
TCM	12 meridians	12 spectral leaves	Circulation channels
TCM	Pulse diagnosis	Spectral dashboard	Flow measurement
TCM	Acupuncture	Activation of thresholds P_4, N_4	Blockage treatment
<i>Tianming</i>	Celestial Empire	System conforming to Heaven ($\mathcal{F}_{Stase} < 0.3$)	Cosmic legitimacy
<i>Tianming</i>	Loss of Mandate	Retropolarity ($R_{seuil} > 0.7$)	Collapse
<i>Tianming</i>	New dynasty	Transition via N_4 (TRR)	Regeneration

Chapter 20 – The geopolitical impact of the TRR: the end of spatial avoidance

20.1. Reduction of dependence on exports: the TRR as a substitute for economic imperialism

The West temporarily “hijacked” the fatal cycle of the Nodal Point through spatial imperialism (colonies, subsoil) and temporal avoidance (financialization). Post-1978 China used spatial avoidance (globalization, construction) and temporal avoidance (LGFV debt). In 2018, the external material frontier is practically exhausted, pushing China to relocate the destruction of its wealth via the SECC (computing overcapacity). The TRR offers a strategic alternative to this flight forward. By recreating endogenous Value on national territory, China structurally reduces its dependence on immaterial exports and economic imperialism. The TRR acts as an internalized substitute for geographic expansion.

20.2. Redefinition of the New Silk Roads: from spatial expansion to the diplomacy of regeneration

The relative failure of productive investments abroad (HPAL factories in Indonesia, Venezuela) proved China’s vulnerability to the **Sovereign Rent** of host states. The strategy reoriented towards “Hardened Enclaves” ports (Chancay, Piraeus, Hambantota) to secure raw material flows. However, these enclaves do not generate the “advanced exogenous demand” necessary to restart the Chinese engine. The TRR allows redefining the *Belt and Road* initiative (BRI): instead of exporting an extraction model (logistical colonization), China can export a **topological regeneration model**. Developing countries, also blocked at their Nodal Point, will be able to import the operator T_{mixed} (Two Mountains Bank, CUR) to trigger their own ecological Extraction cycle.

20.3. Sino-American rivalry: the asymmetric advantage of endogenous regeneration over Western financialization

The Western economy has been in a similar topological dead-end since the 1970s, but has responded with financialization (fictitious bubbles) and massive monetization (dilutive inflation). The TRR offers China a decisive asymmetric advantage:

- **The West attempts direct transitions** ($C_N \rightarrow C_P$) through liquidity injections, violating nilpotence, which prepares a global systemic retropolarity.
- **China, through the TRR, activates the operator T_{mixed}** which respects the conservation law by creating a new underlying asset (regenerative value). By mastering the topologically authorized trajectory, China positions the *Tiandao Jingji* (the Economic Heavenly Way) as the new standard of regulation of complex systems, offering a viable civilizational model facing the collapse of the continuous liberal model.

Table 8: Geopolitical comparison of Nodal Point exit strategies

Strategy	Western model (Financialization)	Post-2018 China (SECC/Enclaves)	Post-TRR China (T_{mixed})
Topological mechanism	Forced direct transition $C_N \rightarrow C_P$ (violation of nilpotence)	External spatial avoidance (Hardened Enclaves) & temporal (SECC)	Controlled crossing through threshold N_4
Underlying asset	Fictitious currency (Debt bubbles)	Immaterial tokens / Mineral ports	Territorial GEP (Regeneration)
Social impact	Destruction of industrial jobs, inequalities	Algorithmic Stasis, <i>Tang Ping</i>	Rural mass employment, restoration of <i>De</i>
Geopolitical posture	Monetary imperialism & sanctions	Logistical imperialism & Sovereign Rent	Diplomacy of regeneration (export of TRR model)
Expected spectral outcome	Retropolarity (Terminal collapse)	Prolonged Stasis (Slow decline)	Healthy growth ($\mathcal{F}_{\text{Stase}} < 0.3$)

Conclusion of Part VI: The stake of the TRR goes beyond China’s borders. By demonstrating that it is possible to exit the Nodal Point without massive wealth destruction, China offers the world the first operational application of a physics of regulation. The Cl(6,6) formalism, the mathematized translation of Chinese structural wisdom, becomes a geopolitical tool of the first order, capable of rallying the Global South by offering it a finally sustainable path of development.

GENERAL CONCLUSION

The structural crisis that China has been going through since 2018 is neither a cyclical accident, nor a short-term governance error. It is the topological expression of the attainment of the **terminal Nodal Point**: the simultaneous exhaustion of spatial avoidance mechanisms (saturation of the territory by construction and mega-infrastructure) and temporal avoidance (asphyxiation by LGFV debt). The “Third Mutation” towards the immaterial (AI, carbon, data), due to its structural deficit of spatial thickness and its derisory employment elasticity (0.14), failed to restart the exogenous demand engine. The system is in **Stasis** (blockage on attractor S), and society translates this topological blockage through the symptom of *Tang Ping*.

The topological model Cl(6, 6) rigorously demonstrates that any attempt to exit the crisis by classic paths (monetary stimulus, forced technological leap) is equivalent to a direct transition between the recession belt (C_N) and the expansion belt (C_P) without crossing the polar threshold N_4 . Such a violation of nilpotence would mechanically trigger a **retropolarity**, that is, a total systemic collapse. Without intervention, China is heading straight for the **tenth dynastic decline** of its history, calibrated on the analysis of the nine previous collapses.

The only topologically authorized trajectory to avoid this massive destruction of wealth is the activation of a mixed transition operator, T_{mixed} . This operator is the **Territorial Regeneration Rent (TRR)**. By substituting

the exhaustion of virgin lands with the qualitative regeneration of degraded spaces, the TRR creates a new Space of Value. It restores the circulation of economic Qi ($\eta > 0$), restores the harmony of Wuxing, and absorbs the fracture between the material and cognitive sectors. Already empirically at work in Lishui, Sanming, or Hainan, it now requires an orchestrated national deployment.

For the public decision-maker, the CI(6,6) formalism offers much more than a critical theory: it provides a **spectral dashboard** at five levels of resolution. Articulated around the composite Stasis index ($\mathcal{F}_{Stase}$), it allows quantifying the Mandate of Heaven, anticipating tipping points, and precisely calibrating the activation of the three structural reforms (Two Mountains Bank, GEP in KPIs, CUR) to sequentially cross thresholds S_5 , S_6 , and S_7 .

The strategic window of the **15th Five-Year Plan (2026-2030)** is unique. In succeeding in this transition, China will not only avoid the tenth decline: it will become the first state in the world to possess an operational physics of regulation. Exported diplomatically to countries of the Global South also blocked at their Nodal Point, this “diplomacy of regeneration” will replace logistical imperialism, offering the Economic Heavenly Way (*Tiandao Jingji*) the status of a new viable civilizational standard.

Table 9: Operational synthesis: The 3 TRR reforms as operator T_{mixed}

Structural reform	Component	Targeted threshold	Topological and economic effect
Two Mountains Bank	T_{water}	S_6 (Water)	Securitization of GEP to circumvent LGFV debt. Reorientation of capital flows towards ecological regeneration.
GEP in local KPIs	$T_{\text{structure}}$	S_5 (Fire)	Reform of political incentives. Institutional anchoring of the new territorial productive model.
Regenerative Usufruct Certificates (CUR)	T_{fire}	S_7 (Octave jump)	Reappropriation of rent by communities. Creation of a new value distribution cycle.

Table 10: 2032 spectral simulation: Trajectories with and without TRR

Indicator (Horizon 2032)	Without TRR (Status quo)	With TRR (Plan 2026-2030)	Recovery Delta
η_{total} (Direction of Qi)	-0.80	+0.35	+1.15
d_{total} (Structural harmony)	0.8	4.2	+3.4
gap (Shock robustness)	0.02	0.45	+0.43
R_{seuil} (Systemic tension)	0.95	0.25	-0.70
$\mathcal{T}_{\text{Wuxing}}$ (Imbalance)	1.4	0.3	-1.1
$\mathcal{F}_{\text{Stase}}$ (Global index / Mandate)	0.80 (<i>Retropolarity</i>)	0.22 (<i>Healthy growth</i>)	-0.58

APPENDICES (Strategic Selection)

The attached appendices constitute a selection of the dashboards and glossaries essential to the understanding of the model for the public decision-maker. The 228-page source document contains all the exhaustive reference tables (notably the complete cartography of the 144 pentads and the matrix of the 64 historical states), available in the separate technical document for use by modeling teams.

Appendix A – Glossary of 15 technical terms (decision-maker)

This glossary provides the political translation of topological concepts for public decision-makers.

Table 11: Topological steering glossary

Term	Economic and political equivalent
Attractor	Stable economic regime (e.g.: polarized growth, recession, depression).
Tropical belt (C_P/C_N)	Cycle of economic policies (expansion or recession).
η (Spectral asymmetry)	The "Qi" of the system: indicates expansion (> 0) or recession (< 0).
d (Spectral dimension)	The harmony of the system: diversity (≈ 5) or excessive fragmentation (< 2).
gap (Spectral)	The margin of maneuver: robustness against exogenous shocks.
LGFV	Local Government Financing Vehicle (instrument of land debt).
Mandate of Heaven	Legitimacy of power, quantified by the Stasis index $\mathcal{F}_{Stase} < 0.3$.
Nilpotence	Condition of systemic stability: its violation causes collapse.
Nodal Point	Threshold of exhaustion of the easily mobilizable Value stock (end of the growth engine).
Extraction	Transfer of a "frozen" asset (land) towards the productive circuit by anticipatory credit.
TRR	Territorial Regeneration Rent: ecological transition operator exiting Stasis.
Retropolarity	Systemic collapse caused by a direct transition without passing through a threshold.
R_{seuil} (Tension)	Tension on the system's gates: red alert if > 0.7 .
SECC	Computing Capacity Export Strategy (external immaterial palliative).
Stasis	Absorbing attractor: state of blockage where the destruction of wealth no longer recreates value.

Appendix B –The 20 attractors: pentad triplets, IDs, economic interpretation and Chinese examples

Table 12: The 20 attractors, their pentad triplets, polarity, IDs, economic interpretation and Chinese examples

Attractor	Pentad triplet	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
A	$\{P_1, P_2, P_4\}$	3P	4	0, 1, 2, 3	Extreme polarized growth: inflation + high profits, extraction blocked. Point of maximum accumulation before rupture.	Post-An Lushan Tang (758-763): inflation > 5000%, rice price from 13 to 1000 wen/dou; Late Southern Song: hyperinflation by over-issuance of <i>jiazi</i>
B	$\{P_1, P_3, P_5\}$	3P	3	4, 5, 6	Export-oriented polarized growth: inflation, high profits, dispersion. Dependence on external markets.	Ming Longqing opening (1567-1644): influx of silver from Japan and America, moderate inflation, massive exports of silk and porcelain
C	$\{P_2, P_3, P_6\}$	3P	4	8, 9, 10, 11	Polarized growth with spatial dispersion: financial bubbles, systemic fragility.	Song (960-1279): commercial boom, <i>jiaozi</i> (paper money), chronic inflation by over-issuance; differentiated regional economic development
D	$\{P_4, P_5, N_2\}$	2P+1N	4	12, 13, 14, 15	Tempered growth: balanced investment and consumption, active extraction.	Wen-Jing rule (180-141 BC): price drop, "a bushel of millet for 10 to 20 wen"; Kaiyuan flourishing (713-741): "a bushel of rice for 3 wen", prosperity, tax reduction
E	$\{P_5, P_6, N_3\}$	2P+1N	4	16, 17, 18, 19	Urban expansion: growth driven by urbanization, low wages, moderate profits.	Northern Song urban commercial prosperity (960-1127): rapid urbanization, growing urban population, moderate inflation by <i>jiaozi</i> issuance
F	$\{P_1, P_6, N_4\}$	2P+1N	3	21, 22, 23	Emerging stagflation: inflation without real wage growth, nodal point.	Mid-late Ming (16th c.): influx of silver, "silver abundant, goods expensive", emerging stagflation

Continued on next page...

The 20 attractors (continued)

Attractor	Pentad triplet	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
G	$\{P_2, P_5, N_6\}$	2P+1N	3	25, 26, 27	Impoverished growth: rising GDP, but stagnant purchasing power, nodal extraction.	Mid-Qing (18th c.): rapid population growth, constant rice price rise, declining peasant purchasing power
H	$\{P_3, P_4, N_6\}$	2P+1N	3	29, 30, 31	Rentier economy: profits from property and monopolies.	Late Ming: land accumulation by elites, land concentration, peasants burdened with debt, revolts
I	$\{P_1, N_2, N_6\}$	1P+2N	3	33, 34, 35	Deflationary recession: falling prices, purchasing power maintained.	Late Ming silver crisis (1630-1644): drop in silver imports, severe deflation, price collapse
J	$\{P_1, N_3, N_5\}$	1P+2N	3	37, 38, 39	Liquidity crisis: deflation + low profits, cascading bankruptcies.	Late Tang (9th c.): monetary crisis, copper shortage, poor quality coinage, exchange chaos
K	$\{P_2, N_3, N_5\}$	1P+2N	4	40, 41, 42, 43	Agricultural depression: sectoral shock, rural exodus.	Late Ming Little Ice Age (17th c.): crop failures, famines, collapse of agricultural production, peasant revolts
L	$\{P_3, N_2, N_4\}$	1P+2N	3	45, 46, 47	War economy: inflation + nodal extraction, productive destruction.	Three Kingdoms period (220-280): fragmentation, regional war economies, inflation, requisitions
M	$\{P_4, N_1, N_3\}$	1P+2N	3	49, 50, 51	Nodal point in frontier: exhaustion of available resources.	Qing territorial expansion (17-18th c.): annexation of Xinjiang, Tibet, Mongolia, frontiers, but high costs
N	$\{P_4, N_5, N_6\}$	1P+2N	3	53, 54, 55	Nodal point with dispersion: territorial fragmentation.	Late Tang (9-10th c.): power fragmentation, autonomous regional economies, decline of interregional trade

Continued on next page...

The 20 attractors (continued)

Attractor	Pentad triplet	Pol.	Deg.	IDs (0-based)	Economic interpretation	Chinese examples
O	$\{P_5, N_1, N_4\}$	1P+2N	3	57, 58, 59	Frontier in deflation: unprofitable colonization.	Han westward expansion (2nd c. BC): colonization of Western regions, high costs, uncertain returns
P	$\{P_6, N_1, N_2\}$	1P+2N	2	61, 63	Dispersion in ecumene regime: chosen de-urbanization, retreat.	Southern Song retreat (1127): mass exodus South, dispersion of population and activities, economic reorganization
Q	$\{P_2, N_1, N_4\}$	1P+2N	3	6, 20, 24	Recession with low purchasing power: austerity, social crisis.	Sui-Tang transition (7th c.): collapse of Sui, wars, famines, collapse of purchasing power
R	$\{P_3, N_1, N_5\}$	1P+2N	3	28, 32, 36	Stagflation: inflation + high profits, nodal extraction.	Song-Jin wars (12-13th c.): inflation by paper money issuance, economic stagnation due to wars
S	$\{P_6, N_5, N_6\}$	1P+2N	3	44, 48, 52	Dispersion and nodal point: systemic fragmentation.	Late Qing (19th c.): Opium Wars, unequal treaties, economic fragmentation, collapse of monetary system
T	$\{N_2, N_3, N_4\}$	3N	3	56, 60, 62	Terminal depression: deflation, low profits, nodal point, wealth destruction.	Late Ming (1644): combination of deflation, famines, peasant revolts, Manchu invasion; Late Qing (1911): terminal crisis, collapse of dynasty

Appendix C – The administrative hierarchy of the model

Table 13: Multi-level steering architecture: the four decision-making layers

Level	Scale	Observables used	Typical action	Horizon
Strategic (Beijing)	National	4 global (η, d, gap, R_{seuil})	Activate TRR if $R_{seuil} > 0.7$; monetary reform if $gap < 0.2$	5–10 years
Tactical (Province)	Provincial	48 per leaf ($\eta_f, d_f, gap_f, R_{seuil,f}$)	Adjust GEP according to η_f ; reorient CURs	1–3 years
Operational (County)	County	144 pentads ($a_{k,g}, c_{k,g}, E_{k,g}$)	Target CUR on active pentads; pilot projects	6–18 months
Diagnostic (Center)	Transversal	7 thresholds (σ_i) + 66 relational	Detect pre-retropolarity; early warning	Real-time

Appendix D – The 7 spectral thresholds and their political significance

Table 14: The 7 spectral thresholds of economic regulation

Threshold	Value $\sqrt{\lambda_i}$	Degeneracy	Economic role	Associated policy
S_1	0.06614	2	Minimal activation	Inflation threshold: start of central bank reaction
S_2	0.10486	2	Mirror symmetry	Tipping between expansion (Sheng) and recession (Ke)
S_3	0.10582	1	Polarization	Policy change
S_4	0.13530	2	Inter-sectoral coupling	Macroeconomic coordination
S_5	0.17195	1	Fire Transition (exogenous shock)	Crisis management, innovation (GEP in KPIs)
S_6	0.19329	1	Water Transition (monetary policy)	Structural reforms (Two Mountains Bank)
S_7	0.21732	1	Octave jump	Cycle change (CUR, generalization)

Appendix E – Mapping of TRR pilot zones

Table 15: Detailed mapping of TRR pilot zones

Zone	Official Chinese name	Degraded Wealth extracted	TRR mechanism
Lishui (Zhejiang)	丽水”两山银行”	Impoverished agricultural lands, abandoned villages.	Two Mountains Bank: aggregation of use rights, loans backed by GEP.
Sanming and Nanping (Fujian)	林票制度 / 森林生态银行	Fragmented collective forests, without market value.	State-owned enterprises buy management rights; villagers receive ”Lin-piao”.
Xingguo and Ganzhou (Jiangxi)	山水林田湖草生态保护修复工程	Eroded watersheds, soils contaminated by rare earth mines.	Ecosystem service payments between cities (upstream/downstream).
Hainan	国家生态文明试验区 / GEP 核算体系	Island territory threatened by real estate speculation.	Ban on speculative construction; GEP as bank collateral.
Huzhou (Zhejiang) and Chongyi (Jiangxi)	竹林碳汇 / 林业碳票	Aging and unharvested bamboo groves.	TRR / Third Mutation bridge: carbon credit exchange on national market.

Appendix F – Detailed simulation of the two scenarios (with/without TRR)

Table 16: 2026-2032 spectral simulation: Trajectories with and without TRR

Scenario 1: Without TRR (Status quo)	2026	2028	2030	2032	Diagnosis 2032
η_{total}	-0.45	-0.50	-0.60	-0.80	Total collapse of Qi
d_{total}	1.9	1.8	1.2	0.8	Extreme fragmentation
gap	0.10	0.08	0.05	0.02	Absolute fragility
R_{seuil}	0.78	0.82	0.90	0.95	Critical saturation
$\mathcal{F}_{Stase}$	0.64	0.70	0.75	0.80	Retropolarity (10th decline)
Scenario 2: With TRR (Plan 2026-2030)	2026	2028	2030	2032	Diagnosis 2032
η_{total}	-0.45	-0.20	+0.10	+0.35	Qi circulates abundantly
d_{total}	1.9	2.8	3.5	4.2	Harmony and diversity
gap	0.10	0.25	0.35	0.45	System robustness
R_{seuil}	0.78	0.55	0.35	0.25	Exit from red zone
$\mathcal{F}_{Stase}$	0.64	0.48	0.32	0.22	Healthy growth

Appendix G – The 9 Chinese dynastic declines according to Rent Extraction Theory

Table 17: The 9 Chinese dynastic declines – Centuries-long trends and crisis mechanisms

Dynasty	Initial structural Mandate	Extraction / Avoidance phases	Crisis mechanism (trend exhaustion)	Example of destruction
Han (206 BC – 220)	End of slavery, point cadastre, closure of the frontier. Shift of value towards land. The State, through price control (the <i>pingzhun</i> system), attempted to regulate the market, but the concentration of land by great families gradually undermined this mechanism.	Initial trend: growth based on free small peasantry. Spatial avoidance limited by the saturation of arable land.	Exhaustion of the first land extraction trend. The State can no longer control prices in the face of appropriation by large landowners, preventing wage regradation. The fiscal crisis and the inability to finance the army led to the weakening of central power.	Yellow Turban revolts (184), civil war, fragmentation into Three Kingdoms.
Fragmentation / 5 Barbarians (220 – 581)	Collapse of the central state. Nomadic invasions (<i>Wu Hu</i>) confiscate the lands of the North and reduce the peasantry to serfdom. Flight of elites and new frontier in the South.	Spatial avoidance (South): development of new southern lands. Servile mode (North): capture of populations to supply labor. The Northern regimes adopt a system of military-agricultural colonies (<i>tuntian</i>).	The conquerors of the North sinicize and adopt land rent. The servile mode exhausts itself. In the South, the frontier closes rapidly under demographic pressure. The monetary system collapses, replaced by barter and silk.	Endless wars, demographic collapse, violent reunification under the Sui (581).
Tang (618 – 907)	<i>Jun-tian</i> (equal fields) system, restoration of small peasantry, westward territorial expansion. The State redistributes land and controls the population through the census.	Spatial avoidance: conquests in Central Asia, Silk Road. The <i>Jun-tian</i> system redistributes land but saturates under demographic pressure. The tax system evolves from labor corvée to a tax in kind and then in money.	Failure of the spatial avoidance of the <i>Jun-tian</i> system. Demographic pressure saturates the frontier, and price rigidity causes a double scarcity (land/labor). The regional lords (<i>jiedushi</i>) appropriate lands and create autonomous military fiefdoms.	An Lushan rebellion (755-763), decline of central power, rise of warlords.

Continued on next page

The 9 Chinese dynastic declines (continued)

Dynasty	Initial structural Mandate	Extraction / Avoidance phases	Crisis mechanism (trend exhaustion)	Example of destruction
Song (960 – 1127)	Maximum commercial development, fiduciary currency (<i>jiaozi</i>), absence of terrestrial frontier. The State relies on maritime trade and monetization of the economy.	Temporal avoidance: monetary and commercial development to compensate for the absence of frontier. Intensive maritime trade, development of banking circuits and credit instruments (<i>huizi</i> , <i>jiaozi</i>). The <i>jiaozi</i> (first paper money, 1024) and later the <i>huizi</i> were massively over-issued to finance military expenditures against the Jurchens and the Mongols.	Temporal avoidance pushed to its paroxysm. Money does not create land. The State can no longer regrade prices in the face of monetary depreciation and land concentration. The fiscal pressure on the peasantry increases, fueling social revolts. The hyperinflation of the <i>huizi</i> (losing 90% of its value between 1200 and 1260) destroyed the purchasing power of the urban middle class.	Jurchen invasions, fall of Kaifeng (1127), loss of Northern China.
43 Jin (North) & Song (South) (1127 – 1279)	Split of the empire. The Jurchen confiscate the lands of the North and impose a semi-servile system (<i>meng'an/mouke</i>). The Southern Song survive on the southern half.	North: reactivation of the servile mode by conquest. South: hyper-specialization of temporal avoidance (credit, massive maritime export). The Southern Song develop a sophisticated monetary economy based on fiduciary money.	Exhausted duality. In the North, military-agricultural stasis. In the South, debt and fiduciary money reach their limits. The system can no longer regulate the overproduction crisis. Inflation becomes chronic.	Hyperinflation in the South, Mongol conquest (1234 in the North, 1279 in the South).
Yuan (1279 – 1368)	Unification under Mongol domination, reintroduction of a frontier by conquest. The Mongols establish a continental empire and reopen the Silk Roads.	Forced spatial avoidance: the Mongol empire opens new territories, but ethnic discrimination blocks redistribution. The monetary system is based on paper money backed by silver.	The Mongol expansion trend exhausts itself rapidly. The Yellow River floods (1340s) and the Black Death aggravate the crisis. The State can no longer control food prices. The monetary system collapses.	Red Turban revolts (1351-1368), founding of the Ming dynasty by Zhu Yuanzhang.

Continued on next page

The 9 Chinese dynastic declines (continued)

Dynasty	Initial structural Mandate	Extraction / Avoidance phases	Crisis mechanism (trend exhaustion)	Example of destruction
Ming (1368 – 1644)	Massive redistribution of land after the war, restoration of small peasantry, the Great Wall as the limit of the frontier. The State prohibits maritime trade (<i>haijin</i>) and fixes taxes in silver.	Initial spatial avoidance: expansion towards the South and Southwest. Then rigidification: maritime prohibition (<i>haijin</i>), taxation fixed in silver. The State relies on the influx of Japanese and American silver via Manila (the "Manila Galleon" trade, 1565-1815). At its peak, 1/3 of the silver mined in Potosí (Bolivia) flowed through Manila to Fujian.	Failure of the initial redistribution. Spatial avoidance stops (closed border). State control of prices can no longer mask massive peasant indebtedness. Spanish silver (via Manila) causes deflationary shocks when the silver supply dries up (1630s crisis: the Spanish crown suspends shipments, causing a 50% drop in silver imports). The Little Ice Age (1620-1700) compounds the crisis with crop failures.	Li Zicheng revolt (1644), fall of Beijing, Manchu invasion.
Qing (1644 – 1912)	Last great territorial expansion (Xinjiang, Tibet, Taiwan), population tripled (100M → 400M). The Qing adopt a policy of ethnic cohabitation and administrative segmentation.	Maximal spatial avoidance: conquest of Central Asia. Then temporal avoidance: unequal treaties, forced opening of ports, external debt. The State attempts to modernize (Self-Strengthening Movement, 1861-1895: creation of the Jiangnan Arsenal, the Fuzhou Shipyard, the Beiyang Fleet) but fails due to institutional rigidity and corruption. The indemnities imposed by the Western powers (450 million taels after the Boxer Rebellion, 1901) strangle the imperial budget.	The centuries-old trend of territorial expansion reaches its geographic limit. Spatial avoidance becomes impossible. Demographic pressure explodes. The revolts (Taiping, 1850-1864: 20-30 million dead; Boxers, 1900) exhaust the imperial finances. Foreign debt strangles the State. The failure of the Hundred Days' Reform (1898) and the late constitutional reforms (1906-1911) come too late to save the dynasty.	Boxer revolt (1900), Xinhai revolution (1911), abdication of the last emperor (1912).

Continued on next page

The 9 Chinese dynastic declines (continued)

Dynasty	Initial structural Mandate	Extraction / Avoidance phases	Crisis mechanism (trend exhaustion)	Example of destruction
Republic (1912 – 1949)	Attempt at modernization, coastal industrialization, collapse of the imperial system. The Republic tries to establish a modern banking system and a unified currency.	Absence of an extraction mechanism: neither frontier nor effective redistribution. Civil war, Japanese occupation, hyperinflation. The State is incapable of controlling prices.	Total absence of an extraction or avoidance mechanism. The market does not function, the State is powerless to regrade prices/wages, leading to the collapse of the price system.	Hyperinflation (monthly rate > 50%), capital flight, Communist victory (1949).

Synthesis: The 9 declines share the same deep structure:

1. A **centuries-long growth trend** initiated by a redistribution (land, frontier, money).
 2. **Avoidance mechanisms** (spatial or temporal) that postpone the crisis.
 3. A **terminal exhaustion** of these mechanisms, aggravated by price rigidity imposed by the State.
 4. A **dynastic change** that resets the system and initiates a new trend.
-

Appendix H – Genesis of the model: from Clifford algebras to complexity invariants (technical version)

This appendix aims to provide modelers with the epistemology and mathematical framework of the model. The objective is to understand how the Clifford algebra $Cl(6, 6)$ articulates with the Rent Extraction Theory (TdR) to produce a universal topological invariant ($64 \rightarrow 20$) applicable to political economy.

1. From Rent Extraction Theory to the discretization of the economy

Classical macroeconomics (DSGE models) postulates a continuous space where variables adjust towards equilibrium. TdR (Rebour, 2000) postulates on the contrary that economic growth is a discrete phenomenon of rupture: “Extraction” (transfer of assets from the Space of Value to the Space of Wealth by monetary anticipation). To model this discreteness, we resort to Clifford algebras. The space of economic configurations $\mathcal{E} = \{0, 1\}^6$ generates $2^6 = 64$ possible states, isomorphic to the 64 codons of the genetic code or the 64 hexagrams of the *Yi Jing*.

2. The algebra $Cl(6, 0)$ and filtration by nilpotence (Rowlands / De Dominicis)

The Clifford algebra $Cl(6, 0)$ is generated by 6 orthogonal elements (the 6 binary economic variables). Peter Rowlands showed that this algebra decomposes into 12 “pentads” (sets of 5 elements: 3 structure bivectors, 1 axial “Fire” element, 1 polar “Water” element). The stability condition of a system is its **nilpotence**: $(g \cdot x)^2 = 0$. If an economic configuration violates this law (for example, hyperinflation coupled with a collapse in production), it is topologically “frustrated” and forbidden. The filtration of the 64 states by the neighborhood rule (geometry of the Merkabah) eliminates these frustrated states and reduces the space to **20 stable attractors**. This is the universal invariant $64 \rightarrow 20$.

3. Extension to $Cl(6, 6)$: Sheng/Ke duality and Λ_{72} lattice

The extension to $Cl(6, 6)$ introduces 6 “anti-cosmic” generators corresponding to regulation variables (Ke mode, contraction) added to the 6 “cosmic” generators (Sheng mode, expansion). This generates $12 \times 12 = 144$ projected pentads. The resonance frequencies of these 144 pentads are encoded in the exceptional lattice Λ_{72} (discovered by Gabriele Nebe). The 48 non-degenerate roots of Λ_{72} provide the critical eigenvalues of the system. The 7 spectral thresholds (S_1 to S_7) used in the steering dashboard are directly extracted from the square roots of these eigenvalues. The discrete Dirac operator $D(t)$ constructed on this lattice allows the calculation of the 4 global spectral observables ($\eta, d, \text{gap}, R_{\text{seuil}}$).

Appendix I – Dashboard user manual (for technical teams)

This manual describes the functional architecture and the calculation algorithm allowing the think tank’s technical teams (or the State Council’s) to feed and interpret the multi-level spectral dashboard.

1. Functional architecture and data ingestion

The dashboard is based on a 5-layer architecture, from the acquisition of raw macroeconomic data to political recommendation.

Table 18: Functional architecture of the spectral steering application

Layer	Content and processing
1. Data sources	Ingestion of macro data (NBS), financial (LGFV debt, credit), territorial (land prices, GEP) and sectoral. Min-max normalization.
2. Pentad extraction	Projection of normalized data onto the $Cl(6, 6)$ space. Identification of the binary vector $\mathbf{x} \in \{0, 1\}^6$. Extraction of the active pentad triplet.
3. Spectral calculation	Construction of the Dirac operator $D(t)$. Resolution of the eigenvalue problem. Calculation of the 4 global observables and the 48 local observables.
4. Dashboard	Generation of gauges, Beijing radar, heat map and correlation matrix. Update of the threshold vector $\vec{\sigma}$ and the index $\mathcal{F}_{Stase}$.
5. Recommendations	Mapping of spectral states to intervention protocols. Activation of alerts (traffic lights) and generation of public policy templates (e.g.: TRR activation).

2. Observable calculation algorithm

The following algorithm describes the processing sequence for a time step t (e.g.: quarter 2026).

Table 19: Algorithm for calculating spectral observables (time step t)

Step	Description of processing
1	Normalization of data $X(t)$ to obtain $\varphi_v(t)$ (activation rate) for each pentad $v \in V$.
2	Determination of the binary configuration $\mathbf{x}(t)$ from macroeconomic thresholds.
3	Identification of the attractor $A_m(t)$ via the projection operator $\Pi(\mathbf{x})$.
4	Extraction of the active pentad triplet $\{P_a, P_b, P_c\}$ and their local polarities $\varepsilon_v \in \{+1, -1\}$.
5	Construction of the adjacency matrix A of the dual graph Γ and calculation of coupling weights $w_{ij}(t)$.
6	Construction of the discrete Dirac operator $D(t)$ (12×12 matrix per spectral leaf).
7	Resolution of the spectral problem: $D(t)\psi = \lambda\psi$. Sorting of ordered eigenvalues: $\lambda_1 \leq \dots \leq \lambda_{12}$.
8	Calculation of asymmetry: $\eta(t) = \frac{1}{\Lambda} \sum sgn(\lambda_k) \lambda_k ^\alpha$.
9	Calculation of dimension: $d(t) = \exp(-\sum p_k \ln p_k)$ with $p_k = \lambda_k / \sum \lambda_j $.
10	Calculation of the gap: $\text{gap}(t) = \min\{ \lambda_k(t) > 0\}$.
11	Calculation of tension: $R_{\text{seuil}}(t) = (\lambda_{p_4} + \lambda_{N_4}) / \sum \lambda_k $.
12	Multi-level aggregation, calculation of fragmentation $\mathcal{F}$, Wuxing tension $\mathcal{T}_{\text{Wuxing}}$, and composite index $\mathcal{F}_{\text{Stase}}$.

3. Alert and validation procedure (Example: China 2026 Red Alert)

The technical team must follow the following procedure when a red alert is triggered by the system:

- Integrity verification:** Check the freshness of NBS and Ministry of Finance data. Ensure that the recorded LGFV debt rate justifies the crossing of $R_{\text{seuil}} > 0.7$.
- Multi-level diagnosis:**
 - Level 1:* Confirm $\mathcal{F}_{\text{Stase}} \geq 0.65$ (characterized Stasis).
 - Level 3:* Check the vector $\vec{\sigma}$. If $\vec{\sigma} = (1, 1, 1, 1, 0, 0, 0)$, confirm the blockage before S_5 .
 - Level 4:* Identify pentad N_5 . If $a_{N_5} > 0.7$, confirm the critical node.
- Intervention simulation:** Inject into the model a variation Δ of the operator T_{mixed} (e.g.: deployment of 50,000 ha of CUR). Observe the projected trajectory of the vector $\vec{\sigma}$ and $\mathcal{F}_{\text{Stase}}$ at T+4 (annual horizon).
- Generation of alert report:** Transmit to the State Council the intervention sheet prescribing the activation of the TRR (Two Mountains Bank, GEP integration) to force the crossing of S_5 .
- Real-time monitoring:** Switch to continuous monitoring mode. Update the algorithm on a monthly sliding window.

TECHNICAL ANNEX DOCUMENT

Mathematical Foundations and Architecture of the Cl(6,6) Steering System

Recipients: Technical teams, modelers, data scientists, econometricians. **Objective:** Provide the formal specification of the topological model and the spectral observable calculation algorithm for the computer implementation of the strategic dashboard.

1. Basic Topological Architecture: From Cl(6,0) to Cl(6,6)

The system is based on the fundamental hypothesis that the economy is a discrete information system isomorphic to the Clifford algebra Cl(6, 6).

1.1. Configuration Space and Clifford Algebra

The space of economic configurations is generated by 6 binary variables (Inflation, Purchasing power, Profits, Extraction, Dispersion, Land regime), forming the space $\mathcal{E} = \{0, 1\}^6$ of size $2^6 = 64$. The extension to signature (6, 6) introduces an “anti-cosmic” sector (regulation variables), formalizing the duality between expansion (Sheng mode, $\eta > 0$) and regulation (Ke mode, $\eta < 0$). The algebra Cl(6, 6) is generated by 12 orthogonal elements:

$$\begin{aligned} \{e_1, \dots, e_6\} & \text{ (cosmic sector, signature +)} \\ \{f_1, \dots, f_6\} & \text{ (anti-cosmic sector, signature -)} \end{aligned}$$

Satisfying the anticommutation relations:

$$e_a e_b + e_b e_a = 2\delta_{ab}, \quad f_a f_b + f_b f_a = -2\delta_{ab}, \quad e_a f_b + f_b e_a = 0$$

Strategic reading: The economy is not a continuous flow of prices, but a network of discrete configurations. The first 6 variables define the visible macroeconomic state (the system’s “pulse”), while the 6 “anti-cosmic” variables model the hidden “reservoir” (stock of latent value, implicit debt, regulatory constraints). Steering the economy consists of orienting the transition between these 12 dimensions in a controlled manner.

1.2. The 144 Pentads and Nilpotence

The basic operator of economic regulation is the **pentad** P , an ordered set of 5 algebraic elements:

$$P = \{B_1, B_2, B_3, F, S\}$$

Where $\{B_1, B_2, B_3\}$ are the bivectors (Institutional structure), $F = i'v$ is the axial element (Fire, exogenous shocks), and $S = 1v$ is the polar element (Water, money/capital). By projection onto the 12 spectral partitions (the e_i and f_j), we obtain the Hilbert space $\mathcal{H}_P$ of the **144 economic pentads**:

$$P_k^{(g)} = g \cdot P_k^{\text{base}}, \quad g \in \{e_1, \dots, e_6, f_1, \dots, f_6\}$$

Stability condition (Nilpotence): An economic configuration is stable if and only if it is nilpotent:

$$(g \cdot x)^2 = 0, \quad \forall g \in \mathcal{G}, \forall x \in P$$

Strategic reading: Nilpotence is the “law of gravity” of the economic system. A regime that violates this law (for example, attempting to maintain hyperinflation coupled with a production collapse, or forcing a monetary stimulus without material anchoring) is topologically “forbidden” and leads to collapse (retropolarity).

2. The Exceptional Lattice Λ_{72} and Spectral Thresholds

The transition dynamics between the 20 stable attractors are constrained by the resonance frequencies of the even unimodular lattice Λ_{72} (discovered by G. Nebe), of dimension 72 and minimal norm $\mu = 8$.

2.1. Universal spectral formula

The energy state of an economic attractor is modeled by the spectral formula:

$$\mathcal{E} = \Lambda_{\text{eco}} \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k$$

Where Λ_{eco} is the scaling constant (to be calibrated), $m \in \{0, 1, 2\}$, $\epsilon_k \in \{-1, 0, 1\}$ is the ternary signature, and β_k are the 15 universal constants derived from the 48 non-degenerate roots of Λ_{72} .

Strategic reading: This formula shows that economic regimes are not ideological choices, but topological “energy levels”. The model provides a ranking of the 20 possible regimes.

2.2. The 7 Spectral Thresholds (S_1 to S_7)

The crossing of these thresholds triggers regime transitions. They correspond to the square roots of the critical eigenvalues $\sqrt{\lambda_i}$ of Λ_{72} . The topological tension $\mathcal{T} = \nabla \eta \cdot \nabla R_{\text{seuil}}$ controls their crossing.

Table 20: Critical values of the 7 spectral thresholds for implementation

Threshold	Value $\sqrt{\lambda_i}$	Degeneracy	Economic role (Trigger)
S_1	0.06614	2	Minimal activation (Central bank reaction)
S_2	0.10486	2	Sheng $\leftrightarrow$ Ke switching (Expansion/Recession)
S_3	0.10582	1	Polarization (Policy change)
S_4	0.13530	2	Inter-sectoral coupling (Macro coordination)
S_5	0.17195	1	Fire Transition (Innovation / Exogenous shock)
S_6	0.19329	1	Water Transition (Monetary reform / Credit)
S_7	0.21732	1	Octave jump (Global cycle change)

Strategic reading: These thresholds are the “energy gates” of the system. The model shows that China is currently blocked between S_4 and S_5 . Attempting to force a direct jump to expansion without crossing S_5 (via KPI reform)

and S_6 (via the Two Mountains Bank) would trigger retropolarity. The political schedule (Five-Year Plan) should follow the sequence $S_5 \rightarrow S_6 \rightarrow S_7$.

3. Dirac Operator and Calculation of Global Observables (Level 1)

The Level 1 diagnosis (Beijing Radar) relies on the diagonalization of the discrete Dirac operator $D(t)$, acting on the graph of the 12 base pentads $\Gamma = (V, E)$.

3.1. Construction of the Operator $D(t)$

For each pentad $F \in V$, we define a local spinor $\psi_F = \begin{pmatrix} \varepsilon_F \\ \varphi_F \end{pmatrix}$, where $\varepsilon_F \in \{+1, -1\}$ is the mode (Sheng/Ke) and $\varphi_F \in [0, 1]$ is the normalized activation rate. The 12×12 Dirac matrix is defined by 2×2 blocks:

$$(D\psi)_F = \sum_{G \sim F} w_{FG}(t) \sigma_{FG} \psi_G$$

Where σ_{FG} are Pauli matrices (σ_x for tropical belt edges, σ_z for polar thresholds), and $w_{FG}(t)$ is the coupling weight:

$$w_{FG}(t) = \beta \cdot \exp\left(-\frac{|\varphi_F - \varphi_G| + E(\varphi_F, \varphi_G)}{\gamma}\right) \cdot \Theta(\varepsilon_F, \varepsilon_G)$$

Strategic reading: The Dirac operator is the “medical scanner” of the State Council. The injection of macroeconomic data (debt rates, GEP, unemployment) into this matrix provides the 4 vital observables that replace the traditional GDP-Inflation-Unemployment trio with a topological diagnosis of Stasis.

3.2. Equations of the 4 Global Observables

The resolution of the eigenvalue equation $D(t)\psi = \lambda\psi$ provides 12 ordered eigenvalues $\lambda_1 \leq \dots \leq \lambda_{12}$. The Level 1 observables are calculated as follows:

1. Spectral asymmetry (η) - The Qi:

$$\eta(t) = \frac{1}{\Lambda} \sum_{k=1}^{12} \text{sgn}(\lambda_k) |\lambda_k|^\alpha$$

(Calibration parameters: $\alpha = 1/2$, Λ normalization factor). *Strategic reading:* Indicates whether the economy is expanding ($\eta > 0$) or contracting ($\eta < 0$). This is the vital breath of the system.

2. Effective spectral dimension (d) - Harmony:

$$d(t) = \exp\left(-\sum_{k=1}^{12} p_k \ln p_k\right) \quad \text{with} \quad p_k = \frac{|\lambda_k|}{\sum_{j=1}^{12} |\lambda_j|}$$

Strategic reading: Measures institutional and territorial diversity. A value close to 5 indicates tempered growth (harmony). If $d < 2$, the system is fragmented and the State loses control over regions.

3. Spectral gap (gap) - Robustness:

$$\text{gap}(t) = \min\{|\lambda_k(t)| > 0\}$$

Strategic reading: This is the State's margin of maneuver before collapse. A low gap means that the slightest geopolitical or climatic crisis will cause the system to rupture.

4. Tension on thresholds (R_{seuil}) - Alert:

$$R_{\text{seuil}}(t) = \frac{|\lambda_{P_4}| + |\lambda_{N_4}|}{\sum_{k=1}^{12} |\lambda_k|}$$

Strategic reading: This is the “red light” of the dashboard. If $R_{\text{seuil}} > 0.7$, the system is attempting to force a transition without paying the institutional price. TRR activation becomes imperative to avoid the 10th decline.

4. Hierarchical Extension and Composite Index (Levels 2 to 5)

The multi-level system requires decomposing $D(t)$ into a direct sum of operators restricted to the 12 spectral leaves $\mathcal{F}_g$:

$$D(t) = \bigoplus_{g \in \mathcal{F}} D_g(t)$$

This generates 144 local eigenvalues, allowing the calculation of the 48 leaf observables (Level 2), the 7 binary thresholds σ_i (Level 3), the activity of the 144 pentads $a_{k,g}$ (Level 4), and the 66 inter-leaf correlations $C_{a,b}$ (Level 5).

4.1. Composite Stasis Index ($\mathcal{F}_{\text{Stase}}$)

The Stasis index synthesizes the 5 levels into a single indicator quantifying the “Mandate of Heaven”.

$$\mathcal{F}_{\text{Stase}}(t) = w_1 |\eta_{\text{total}}| + w_2 \frac{1}{d_{\text{total}}} + w_3 e^{-5 \cdot \text{gap}_{\text{total}}} + w_4 R_{\text{seuil}, \text{total}} + w_5 \mathcal{T}_{\text{Wuxing}}$$

Calibration of weights (w_i) on the 9 dynastic declines: The index is calibrated retrospectively on the historiography of the 9 dynastic declines (from the fall of the Han to the 1949 Republic, including the fragmentations of the Five Barbarians and the Jin/Song split).

Table 21: Weighting parameters of the composite Stasis index $\mathcal{F}_{Stase}$ (Calibrated on 9 declines)

Parameter	Value	Justification (Presence of signal in the 9 declines)
w_1	0.23	Asymmetry (η) is the most robust signal (present in 9/9 declines).
w_2	0.18	Fragmentation ($1/d$) is a signal reinforced by periods of schism (present in 7/9 declines).
w_3	0.23	Spectral fragility (gap) is critical (present in 9/9 declines).
w_4	0.21	Tension on thresholds (R_{seuil}) announces retropolarity (present in 8/9 declines).
w_5	0.15	Wuxing imbalance ($\mathcal{T}_{Wuxing}$) is culturally anchored (present in 6/9 declines).

Validation thresholds: $\mathcal{F}_{Stase} < 0.3$ (Healthy growth); $0.65 \leq \mathcal{F}_{Stase} < 0.75$ (Characterized Stasis, Red Alert); > 0.75 (Pre-retropolarity, imminent collapse).

Strategic reading: The index $\mathcal{F}_{Stase}$ is the quantifier of the Mandate of Heaven. It replaces GDP growth as an indicator of regime legitimacy. A score > 0.65 imposes the activation of the TRR at the risk of losing a decade. For China 2026, the index $\mathcal{F}_{Stase} = \mathbf{0.644}$ confirms the state of red alert.

5. Algorithmic Implementation Note

For development teams, the calculation architecture must respect the following execution order at each time step t (e.g.: quarterly update):

1. **Ingestion:** Collect macro data (NBS, PBoC, NDRC) and normalize (min-max) to generate activation vectors $\varphi_v(t)$ for the 12 base pentads.
2. **Projections:** Apply the spectral projectors Π_g to generate the 144 extended pentads and calculate the matrices $D_g(t)$.
3. **Diagonalization:** Calculate the eigenvalues $\lambda_k^{(g)}(t)$ for each leaf. *Target algorithmic complexity: $O(22615)$, executable in under one second on standard infrastructure.*
4. **Aggregation:** Calculate the 5 levels of observables, update the threshold vector $\vec{\sigma}(t)$ and calculate $\mathcal{F}_{Stase}(t)$.
5. **Visualization/Alert:** Trigger traffic lights (Green/Orange/Red) according to the value of $\mathcal{F}_{Stase}$ and generate the intervention report (e.g.: TRR Activation if $\mathcal{F}_{Stase} \geq 0.65$).

REFERENCES AND KEY SOURCES

This report synthesizes interdisciplinary work and relies on recent macroeconomic data. The references are classified by nature to facilitate consultation by the public decision-maker and the researcher.

1. Theoretical and topological foundations

- **Rebour, T.** (2000, 2011). *La théorie du rachat : Géographie, économie, histoire & Le Point Nodal et la loi de conservation de la Valeur*. Éditions Structurales / Publications de la Sorbonne. (Base ontologique du diagnostic de la Stase).
- **Rowlands, P.** (2007, 2014). *Zero to Infinity: The Foundations of Physics*. World Scientific. (Formalisation de l'algèbre de Clifford $Cl(6,0)$ et des pentades nilpotentes).
- **Nebe, G.** (2010). *An even unimodular 72-dimensional lattice of minimum 8*. arXiv:1008.2862. (Découverte du réseau Λ_{72} gouvernant les seuils spectraux de transition).
- **De Dominicis, B.** (2026). *Wuxing and $Cl(6,6)$: 144 pentads for a unified relational physics & The genetic code as a $64 \rightarrow 20$ clifford invariant*. (Extension à $Cl(6,6)$ et isomorphisme avec la pensée corrélative chinoise). DOI: 19661819
- **De Dominicis, B.** (2026). *Arithmetic unification of masses and energies via the exceptional lattice Λ_{72}* , Application du réseau Λ_{72} à l'unification arithmétique des masses. DOI: 21044905
- **Vandermeersch, L.** (2013). *Les deux raisons de la pensée chinoise : Divination et idéographie*. Gallimard. 中国思想的两种理性: 占卜与表意 (跨文化对话平台丛书) (Épistémologie de la raison divinatoire appliquée au tableau de bord spectral).

2. Macroeconomic and sectoral data (China 2026)

- **National Bureau of Statistics (NBS) of China.** (2024-2026). *National Economic Performance & National Migrant Worker Monitoring Survey Report*. (Data on real estate collapse, investment contraction, and job destruction in construction).
- **China Macroeconomic Forum (CMF), Renmin University.** (2026). *Mid-2026 Report: Assessment of the job creation capacity of the New Productive Forces*. (Calculation of employment elasticity of 0.14 for the AI/cloud sector).
- **China Academy of Information and Communications Technology (CAICT).** (2026). *White Paper on China's Computing Power Development Index*. (Statistics on computing overcapacity of 155 EFLOPS and the SECC Strategy).
- **International Monetary Fund (IMF).** (2023). *People's Republic of China: Article IV Consultation*. (Analysis of asphyxiating LGFV debt and local off-balance-sheet debt).
- **CICC Research (China International Capital Corporation).** (2026). *Macro Outlook H2: Domestic multiplier of data centers vs residential construction*. (Analysis of the spatial thickness deficit of the immaterial).
- **CCID Consulting Group.** (2026). *White Paper on the Evolution of China's Computing Power Industry*. (Data on Western energy bottlenecks and gross margins of Chinese inference).

3. Institutional and geopolitical framework

- **NDRC (National Development and Reform Commission).** (2024-2025). *Notifications on TRR pilot zones and the GEP realization mechanism*. (Empirical mapping of Lishui, Sanming, Ganzhou).

- **Elvin, M.** (1973). *The Pattern of the Chinese Past*. Stanford University Press. (Historical calibration of the model on the 9 dynastic declines and the notion of “high-level equilibrium trap”).
 - **Zhang, G.** (2018). *Tracer le chemin dans l'adversité : Chronique de la décision et de la construction des grands projets du siècle*. Renmin Chubanshe. (Operational account of geographic Extraction: West-East Gas Pipeline, Three Gorges).
-

THEORETICAL AND MATHEMATICAL ANNEX DOCUMENT

Foundations, Isomorphisms and Reference Tables of the Cl(6,6) Model

PART A: ADVANCED MATHEMATICAL FOUNDATIONS

1. The Exceptional Lattice Λ_{72} and the Universal Constants β_k

The transition dynamics between the 20 economic attractors is not continuous; it is constrained by the exceptional lattice Λ_{72} (discovered by G. Nebe). This even unimodular lattice, of dimension 72 and minimal norm $\mu = 8$, provides the complete spectrum of resonance frequencies of the system.

The energy of an economic attractor is modeled by the universal spectral formula:

$$\mathcal{E} = \Lambda_{\text{eco}} \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k$$

Where $\Lambda_{\text{eco}} \approx 1.23 \times 10^{-3}$ (in units of GDP per year), $m \in \{0, 1, 2\}$ according to polarity, $\epsilon_k \in \{-1, 0, 1\}$ is the ternary signature of the attractor, and β_k are the 15 universal constants derived from the 48 non-degenerate roots of Λ_{72} . The eigenvalue intervals dictate the classification of regimes:

- $\lambda > 10$: 3P attractors (polarized growth)
- $5 < \lambda \leq 10$: 2P+1N attractors (tempered growth)
- $2 < \lambda \leq 5$: 1P+2N attractors (recession, stagflation)
- $0 < \lambda \leq 2$: 3N attractor (depression)
- $\lambda \leq 0$: Frustrated configurations (forbidden).

Strategic reading: This arithmetic formalism shows that economic regimes are bounded by universal resonance laws. An economy cannot invent a regime of eternal growth (indefinite 3P): topology forces it to fall back into recession regimes (1P+2N) to purge fictitious wealth.

2. The Transition Operator T and its Selection Rules

The transition operator T acts on the Hilbert space of the 144 pentads. It decomposes into four components:

$$T = T_{\text{structure}} + T_{\text{feu}} + T_{\text{eau}} + T_{\text{mixte}}$$

- $T_{\text{structure}}$ acts on the productive model and institutions.
- T_{feu} acts on exogenous shocks and innovation.
- T_{eau} acts on money and capital.
- T_{mixte} couples the three subspaces (the TRR operator).

The operator T is expressed as an exponential angular rotation:

$$T = \exp \left(i \sum_{a < b} \omega_{ab} L_{ab} \right), \quad L_{ab} = -i \left(\theta_a \frac{\partial}{\partial \theta_b} - \theta_b \frac{\partial}{\partial \theta_a} \right)$$

A matrix element of T is non-zero if and only if 5 strict selection rules are respected (conservation of the number of generators, gauge type, chirality, angular momentum, and preservation of nilpotence).

Strategic reading: The operator T models the impact of political reforms. The selection rules show that an isolated reform (e.g.: monetary policy alone, T_{eau}) cannot bring the system out of Stasis. Only the operator T_{mixte} (which simultaneously combines the institutional, monetary, and innovation reforms of the TRR) respects the conservation laws and allows crossing the threshold N_4 . It is the violation of these selection rules that causes collapses (retropolarities).

PART B: EXHAUSTIVE REFERENCE TABLES

Complete Table of the 144 Economic Pentads of Cl(6,6)

The Hilbert space $\mathcal{H}_P$ is generated by the projection of the 12 base pentads onto the 12 spectral partitions $(e_1 \dots e_6, f_1 \dots f_6)$.

Green: positive pentads P_k (Sheng mode, $\eta > 0$); **Red:** negative pentads N_k (Ke mode, $\eta < 0$).

Table 22: Complete Table of the 144 Economic Pentads of Cl(6,6)

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
Block e_1: Cosmic partition 1 (Growth, Innovation)					
1	$P_1^{(e_1)}$	$\{e_1iI, e_1iJ, e_1iK\}$	$e_1i'k$	e_1j	Keynesian growth with massive public investment
2	$P_2^{(e_1)}$	$\{e_1jI, e_1jJ, e_1jK\}$	$e_1i'i$	e_1k	Liberal market with technological innovation
3	$P_3^{(e_1)}$	$\{e_1kI, e_1kJ, e_1kK\}$	$e_1i'j$	e_1i	Productive debt, credit to innovation
4	$P_4^{(e_1)}$	$\{e_1i'Ii, e_1i'Ij, e_1i'k\}$	$e_1i'K$	e_1J	High-tech exports
5	$P_5^{(e_1)}$	$\{e_1i'Ji, e_1i'Jj, e_1i'Jk\}$	$e_1i'I$	e_1K	Nodal point with disruptive innovation
6	$P_6^{(e_1)}$	$\{e_1i'Ki, e_1i'Kj, e_1i'Kk\}$	$e_1i'J$	e_1I	Domestic consumption boosted by innovation
7	$N_1^{(e_1)}$	$\{-e_1iI, -e_1iJ, -e_1iK\}$	$-e_1i'k$	$-e_1j$	Wage austerity with repression of innovation
8	$N_2^{(e_1)}$	$\{-e_1jI, -e_1jJ, -e_1jK\}$	$-e_1i'i$	$-e_1k$	Structural rigidity hindering innovation
9	$N_3^{(e_1)}$	$\{-e_1kI, -e_1kJ, -e_1kK\}$	$-e_1i'j$	$-e_1i$	Deleveraging without innovation
10	$N_4^{(e_1)}$	$\{-e_1i'Ii, -e_1i'Ij, -e_1i'k\}$	$-e_1i'K$	$-e_1J$	Protectionist retreat hindering innovation
11	$N_5^{(e_1)}$	$\{-e_1i'Ji, -e_1i'Jj, -e_1i'Jk\}$	$-e_1i'I$	$-e_1K$	Stasis with innovation blockage
12	$N_6^{(e_1)}$	$\{-e_1i'Ki, -e_1i'Kj, -e_1i'Kk\}$	$-e_1i'J$	$-e_1I$	Internal austerity without innovation
Block e_2: Cosmic partition 2 (Purchasing power, Wages)					
13	$P_1^{(e_2)}$	$\{e_2iI, e_2iJ, e_2iK\}$	$e_2i'k$	e_2j	Thirty Glorious Years: Keynesianism, high wages
14	$P_2^{(e_2)}$	$\{e_2jI, e_2jJ, e_2jK\}$	$e_2i'i$	e_2k	Liberal market with preserved purchasing power
15	$P_3^{(e_2)}$	$\{e_2kI, e_2kJ, e_2kK\}$	$e_2i'j$	e_2i	Debt with high wages
16	$P_4^{(e_2)}$	$\{e_2i'Ii, e_2i'Ij, e_2i'k\}$	$e_2i'K$	e_2J	Exports with high wages

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
17	$P_5^{(e_2)}$	$\{e_2i'Ji, e_2i'Jj, e_2i'Jk\}$	$e_2i'I$	e_2K	Nodal point with high purchasing power
18	$P_6^{(e_2)}$	$\{e_2i'Ki, e_2i'Kj, e_2i'Kk\}$	$e_2i'J$	e_2I	Domestic consumption, high wages
19	$N_1^{(e_2)}$	$\{-e_2iI, -e_2iJ, -e_2iK\}$	$-e_2i'k$	$-e_2j$	Wage austerity despite purchasing power
20	$N_2^{(e_2)}$	$\{-e_2jI, -e_2jJ, -e_2jK\}$	$-e_2i'i$	$-e_2k$	Rigidity with high wages
21	$N_3^{(e_2)}$	$\{-e_2kI, -e_2kJ, -e_2kK\}$	$-e_2i'j$	$-e_2i$	Deleveraging with high wages
22	$N_4^{(e_2)}$	$\{-e_2i'Ii, -e_2i'Ij, -e_2i'k\}$	$-e_2i'K$	$-e_2J$	New Deal: protectionism, high wages
23	$N_5^{(e_2)}$	$\{-e_2i'Ji, -e_2i'Jj, -e_2i'Jk\}$	$-e_2i'I$	$-e_2K$	Stasis despite purchasing power
24	$N_6^{(e_2)}$	$\{-e_2i'Ki, -e_2i'Kj, -e_2i'Kk\}$	$-e_2i'J$	$-e_2I$	Internal austerity with high wages
Block e_3: Cosmic partition 3 (Profits, Accumulation)					
25	$P_1^{(e_3)}$	$\{e_3iI, e_3iJ, e_3iK\}$	$e_3i'k$	e_3j	Keynesianism with high profits
26	$P_2^{(e_3)}$	$\{e_3jI, e_3jJ, e_3jK\}$	$e_3i'i$	e_3k	Liberal market, maximum profits
27	$P_3^{(e_3)}$	$\{e_3kI, e_3kJ, e_3kK\}$	$e_3i'j$	e_3i	Debt with high profits
28	$P_4^{(e_3)}$	$\{e_3i'Ii, e_3i'Ij, e_3i'k\}$	$e_3i'K$	e_3J	Exports with high profits
29	$P_5^{(e_3)}$	$\{e_3i'Ji, e_3i'Jj, e_3i'Jk\}$	$e_3i'I$	e_3K	Nodal point with high profits
30	$P_6^{(e_3)}$	$\{e_3i'Ki, e_3i'Kj, e_3i'Kk\}$	$e_3i'J$	e_3I	Domestic consumption, high profits
31	$N_1^{(e_3)}$	$\{-e_3iI, -e_3iJ, -e_3iK\}$	$-e_3i'k$	$-e_3j$	Wage austerity, falling profits
32	$N_2^{(e_3)}$	$\{-e_3jI, -e_3jJ, -e_3jK\}$	$-e_3i'i$	$-e_3k$	Rigidity, falling profits
33	$N_3^{(e_3)}$	$\{-e_3kI, -e_3kJ, -e_3kK\}$	$-e_3i'j$	$-e_3i$	Deleveraging, falling profits
34	$N_4^{(e_3)}$	$\{-e_3i'Ii, -e_3i'Ij, -e_3i'k\}$	$-e_3i'K$	$-e_3J$	Protectionism with falling profits

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
35	$N_5^{(e_3)}$	$\{-e_3i'Ji, -e_3i'Jj, -e_3i'Jk\}$	$-e_3i'I$	$-e_3K$	Stasis with falling profits
36	$N_6^{(e_3)}$	$\{-e_3i'Ki, -e_3i'Kj, -e_3i'Kk\}$	$-e_3i'J$	$-e_3I$	Internal austerity, falling profits
Block e_4: Cosmic partition 4 (Active extraction)					
37	$P_1^{(e_4)}$	$\{e_4iI, e_4iJ, e_4iK\}$	$e_4i'k$	e_4j	Keynesianism, active extraction
38	$P_2^{(e_4)}$	$\{e_4jI, e_4jJ, e_4jK\}$	$e_4i'i$	e_4k	Liberal market, active extraction
39	$P_3^{(e_4)}$	$\{e_4kI, e_4kJ, e_4kK\}$	$e_4i'j$	e_4i	Debt, active extraction
40	$P_4^{(e_4)}$	$\{e_4i'Ii, e_4i'Ij, e_4i'k\}$	$e_4i'K$	e_4J	Exports, active extraction
41	$P_5^{(e_4)}$	$\{e_4i'Ji, e_4i'Jj, e_4i'Jk\}$	$e_4i'I$	e_4K	Active extraction in frontier
42	$P_6^{(e_4)}$	$\{e_4i'Ki, e_4i'Kj, e_4i'Kk\}$	$e_4i'J$	e_4I	Domestic consumption, active extraction
43	$N_1^{(e_4)}$	$\{-e_4iI, -e_4iJ, -e_4iK\}$	$-e_4i'k$	$-e_4j$	Austerity, extraction blocked
44	$N_2^{(e_4)}$	$\{-e_4jI, -e_4jJ, -e_4jK\}$	$-e_4i'i$	$-e_4k$	Rigidity, extraction blocked
45	$N_3^{(e_4)}$	$\{-e_4kI, -e_4kJ, -e_4kK\}$	$-e_4i'j$	$-e_4i$	Deleveraging, extraction blocked
46	$N_4^{(e_4)}$	$\{-e_4i'Ii, -e_4i'Ij, -e_4i'k\}$	$-e_4i'K$	$-e_4J$	Protectionism, extraction blocked
47	$N_5^{(e_4)}$	$\{-e_4i'Ji, -e_4i'Jj, -e_4i'Jk\}$	$-e_4i'I$	$-e_4K$	Stasis, extraction blocked
48	$N_6^{(e_4)}$	$\{-e_4i'Ki, -e_4i'Kj, -e_4i'Kk\}$	$-e_4i'J$	$-e_4I$	Internal austerity, extraction blocked
Block e_5: Cosmic partition 5 (Dispersion)					
49	$P_1^{(e_5)}$	$\{e_5iI, e_5iJ, e_5iK\}$	$e_5i'k$	e_5j	Keynesianism with dispersion
50	$P_2^{(e_5)}$	$\{e_5jI, e_5jJ, e_5jK\}$	$e_5i'i$	e_5k	Liberal market with dispersion
51	$P_3^{(e_5)}$	$\{e_5kI, e_5kJ, e_5kK\}$	$e_5i'j$	e_5i	Debt with dispersion

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
52	$P_4^{(e_5)}$	$\{e_5i'Ii, e_5i'Ij, e_5i'k\}$	$e_5i'K$	e_5J	Exports with dispersion
53	$P_5^{(e_5)}$	$\{e_5i'Ji, e_5i'Jj, e_5i'Jk\}$	$e_5i'I$	e_5K	Extraction with spatial dispersion
54	$P_6^{(e_5)}$	$\{e_5i'Ki, e_5i'Kj, e_5i'Kk\}$	$e_5i'J$	e_5I	Domestic consumption with dispersion
55	$N_1^{(e_5)}$	$\{-e_5iI, -e_5iJ, -e_5iK\}$	$-e_5i'k$	$-e_5j$	Austerity with concentration
56	$N_2^{(e_5)}$	$\{-e_5jI, -e_5jJ, -e_5jK\}$	$-e_5i'i$	$-e_5k$	Rigidity with concentration
57	$N_3^{(e_5)}$	$\{-e_5kI, -e_5kJ, -e_5kK\}$	$-e_5i'j$	$-e_5i$	Deleveraging with concentration
58	$N_4^{(e_5)}$	$\{-e_5i'Ii, -e_5i'Ij, -e_5i'k\}$	$-e_5i'K$	$-e_5J$	Protectionism, concentration
59	$N_5^{(e_5)}$	$\{-e_5i'Ji, -e_5i'Jj, -e_5i'Jk\}$	$-e_5i'I$	$-e_5K$	Stasis with concentration
60	$N_6^{(e_5)}$	$\{-e_5i'Ki, -e_5i'Kj, -e_5i'Kk\}$	$-e_5i'J$	$-e_5I$	Internal austerity with concentration
Block e_6: Cosmic partition 6 (Land regime, Frontier)					
61	$P_1^{(e_6)}$	$\{e_6iI, e_6iJ, e_6iK\}$	$e_6i'k$	e_6j	Keynesianism, frontier open
62	$P_2^{(e_6)}$	$\{e_6jI, e_6jJ, e_6jK\}$	$e_6i'i$	e_6k	Liberal market, frontier open
63	$P_3^{(e_6)}$	$\{e_6kI, e_6kJ, e_6kK\}$	$e_6i'j$	e_6i	Debt, frontier open
64	$P_4^{(e_6)}$	$\{e_6i'Ii, e_6i'Ij, e_6i'k\}$	$e_6i'K$	e_6J	Exports, frontier open
65	$P_5^{(e_6)}$	$\{e_6i'Ji, e_6i'Jj, e_6i'Jk\}$	$e_6i'I$	e_6K	Extraction in frontier
66	$P_6^{(e_6)}$	$\{e_6i'Ki, e_6i'Kj, e_6i'Kk\}$	$e_6i'J$	e_6I	Domestic consumption, frontier
67	$N_1^{(e_6)}$	$\{-e_6iI, -e_6iJ, -e_6iK\}$	$-e_6i'k$	$-e_6j$	Austerity, frontier closed
68	$N_2^{(e_6)}$	$\{-e_6jI, -e_6jJ, -e_6jK\}$	$-e_6i'i$	$-e_6k$	Rigidity, frontier closed
69	$N_3^{(e_6)}$	$\{-e_6kI, -e_6kJ, -e_6kK\}$	$-e_6i'j$	$-e_6i$	Deleveraging, frontier closed

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
70	$N_4^{(e_6)}$	$\{-e_6i'Ii, -e_6i'Ij, -e_6i'k\}$	$-e_6i'K$	$-e_6J$	Protectionism, frontier closed
71	$N_5^{(e_6)}$	$\{-e_6i'Ji, -e_6i'Jj, -e_6i'Jk\}$	$-e_6i'I$	$-e_6K$	Stasis, frontier closed
72	$N_6^{(e_6)}$	$\{-e_6i'Ki, -e_6i'Kj, -e_6i'Kk\}$	$-e_6i'J$	$-e_6I$	Internal austerity, frontier closed
Block f_1: Anti-cosmic partition 1 (Wage regulation)					
73	$P_1^{(f_1)}$	$\{f_1iI, f_1iJ, f_1iK\}$	$f_1i'k$	f_1j	Wage regulation, Ke mode
74	$P_2^{(f_1)}$	$\{f_1jI, f_1jJ, f_1jK\}$	$f_1i'i$	f_1k	Market regulation, Ke mode
75	$P_3^{(f_1)}$	$\{f_1kI, f_1kJ, f_1kK\}$	$f_1i'j$	f_1i	Debt regulation
76	$P_4^{(f_1)}$	$\{f_1i'Ii, f_1i'Ij, f_1i'k\}$	$f_1i'K$	f_1J	Export regulation
77	$P_5^{(f_1)}$	$\{f_1i'Ji, f_1i'Jj, f_1i'Jk\}$	$f_1i'I$	f_1K	Extraction regulation
78	$P_6^{(f_1)}$	$\{f_1i'Ki, f_1i'Kj, f_1i'Kk\}$	$f_1i'J$	f_1I	Consumption regulation
79	$N_1^{(f_1)}$	$\{-f_1iI, -f_1iJ, -f_1iK\}$	$-f_1i'k$	$-f_1j$	Wage deregulation
80	$N_2^{(f_1)}$	$\{-f_1jI, -f_1jJ, -f_1jK\}$	$-f_1i'i$	$-f_1k$	Market deregulation
81	$N_3^{(f_1)}$	$\{-f_1kI, -f_1kJ, -f_1kK\}$	$-f_1i'j$	$-f_1i$	Debt deregulation
82	$N_4^{(f_1)}$	$\{-f_1i'Ii, -f_1i'Ij, -f_1i'k\}$	$-f_1i'K$	$-f_1J$	Export deregulation
83	$N_5^{(f_1)}$	$\{-f_1i'Ji, -f_1i'Jj, -f_1i'Jk\}$	$-f_1i'I$	$-f_1K$	Extraction deregulation
84	$N_6^{(f_1)}$	$\{-f_1i'Ki, -f_1i'Kj, -f_1i'Kk\}$	$-f_1i'J$	$-f_1I$	Consumption deregulation
Block f_2: Anti-cosmic partition 2 (Financial regulation)					
85	$P_1^{(f_2)}$	$\{f_2iI, f_2iJ, f_2iK\}$	$f_2i'k$	f_2j	Keynesian financial regulation
86	$P_2^{(f_2)}$	$\{f_2jI, f_2jJ, f_2jK\}$	$f_2i'i$	f_2k	Liberal financial regulation

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
87	$P_3^{(f_2)}$	$\{f_2kI, f_2kJ, f_2kK\}$	$f_2i'j$	f_2i	Credit regulation
88	$P_4^{(f_2)}$	$\{f_2i'Ii, f_2i'Ij, f_2i'k\}$	$f_2i'K$	f_2J	Exchange rate regulation
89	$P_5^{(f_2)}$	$\{f_2i'Ji, f_2i'Jj, f_2i'Jk\}$	$f_2i'I$	f_2K	Debt regulation
90	$P_6^{(f_2)}$	$\{f_2i'Ki, f_2i'Kj, f_2i'Kk\}$	$f_2i'J$	f_2I	Savings regulation
91	$N_1^{(f_2)}$	$\{-f_2iI, -f_2iJ, -f_2iK\}$	$-f_2i'k$	$-f_2j$	Financial deregulation
92	$N_2^{(f_2)}$	$\{-f_2jI, -f_2jJ, -f_2jK\}$	$-f_2i'i$	$-f_2k$	Credit deregulation
93	$N_3^{(f_2)}$	$\{-f_2kI, -f_2kJ, -f_2kK\}$	$-f_2i'j$	$-f_2i$	Debt deregulation
94	$N_4^{(f_2)}$	$\{-f_2i'Ii, -f_2i'Ij, -f_2i'k\}$	$-f_2i'K$	$-f_2J$	Exchange rate deregulation
95	$N_5^{(f_2)}$	$\{-f_2i'Ji, -f_2i'Jj, -f_2i'Jk\}$	$-f_2i'I$	$-f_2K$	Debt deregulation
96	$N_6^{(f_2)}$	$\{-f_2i'Ki, -f_2i'Kj, -f_2i'Kk\}$	$-f_2i'J$	$-f_2I$	Savings deregulation
Block f_3: Anti-cosmic partition 3 (Profit regulation)					
97	$P_1^{(f_3)}$	$\{f_3iI, f_3iJ, f_3iK\}$	$f_3i'k$	f_3j	Profit regulation, Ke mode
98	$P_2^{(f_3)}$	$\{f_3jI, f_3jJ, f_3jK\}$	$f_3i'i$	f_3k	Margin regulation
99	$P_3^{(f_3)}$	$\{f_3kI, f_3kJ, f_3kK\}$	$f_3i'j$	f_3i	Investment regulation
100	$P_4^{(f_3)}$	$\{f_3i'Ii, f_3i'Ij, f_3i'k\}$	$f_3i'K$	f_3J	Export profit regulation
101	$P_5^{(f_3)}$	$\{f_3i'Ji, f_3i'Jj, f_3i'Jk\}$	$f_3i'I$	f_3K	Extraction profit regulation
102	$P_6^{(f_3)}$	$\{f_3i'Ki, f_3i'Kj, f_3i'Kk\}$	$f_3i'J$	f_3I	Rent regulation
103	$N_1^{(f_3)}$	$\{-f_3iI, -f_3iJ, -f_3iK\}$	$-f_3i'k$	$-f_3j$	Profit deregulation
104	$N_2^{(f_3)}$	$\{-f_3jI, -f_3jJ, -f_3jK\}$	$-f_3i'i$	$-f_3k$	Margin deregulation

Continued on next page...

Complete Table of the 144 Economic Pentads of $Cl(6,6)$ – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
105	$N_3^{(f_3)}$	$\{-f_3kI, -f_3kJ, -f_3kK\}$	$-f_3i'j$	$-f_3i$	Investment deregulation
106	$N_4^{(f_3)}$	$\{-f_3i'Ii, -f_3i'Ij, -f_3i'k\}$	$-f_3i'K$	$-f_3J$	Export profit deregulation
107	$N_5^{(f_3)}$	$\{-f_3i'Ji, -f_3i'Jj, -f_3i'Jk\}$	$-f_3i'I$	$-f_3K$	Stasis: total deregulation
108	$N_6^{(f_3)}$	$\{-f_3i'Ki, -f_3i'Kj, -f_3i'Kk\}$	$-f_3i'J$	$-f_3I$	Rent deregulation
Block f_4: Anti-cosmic partition 4 (Extraction regulation)					
109	$P_1^{(f_4)}$	$\{f_4iI, f_4iJ, f_4iK\}$	$f_4i'k$	f_4j	Keynesian extraction regulation
110	$P_2^{(f_4)}$	$\{f_4jI, f_4jJ, f_4jK\}$	$f_4i'i$	f_4k	Liberal extraction regulation
111	$P_3^{(f_4)}$	$\{f_4kI, f_4kJ, f_4kK\}$	$f_4i'j$	f_4i	Extraction credit regulation
112	$P_4^{(f_4)}$	$\{f_4i'Ii, f_4i'Ij, f_4i'k\}$	$f_4i'K$	f_4J	N_4: Export extraction regulation
113	$P_5^{(f_4)}$	$\{f_4i'Ji, f_4i'Jj, f_4i'Jk\}$	$f_4i'I$	f_4K	Nodal point regulation
114	$P_6^{(f_4)}$	$\{f_4i'Ki, f_4i'Kj, f_4i'Kk\}$	$f_4i'J$	f_4I	Consumption regulation
115	$N_1^{(f_4)}$	$\{-f_4iI, -f_4iJ, -f_4iK\}$	$-f_4i'k$	$-f_4j$	Extraction deregulation
116	$N_2^{(f_4)}$	$\{-f_4jI, -f_4jJ, -f_4jK\}$	$-f_4i'i$	$-f_4k$	Credit deregulation
117	$N_3^{(f_4)}$	$\{-f_4kI, -f_4kJ, -f_4kK\}$	$-f_4i'j$	$-f_4i$	Debt deregulation
118	$N_4^{(f_4)}$	$\{-f_4i'Ii, -f_4i'Ij, -f_4i'k\}$	$-f_4i'K$	$-f_4J$	$-N_4$: Export deregulation
119	$N_5^{(f_4)}$	$\{-f_4i'Ji, -f_4i'Jj, -f_4i'Jk\}$	$-f_4i'I$	$-f_4K$	Nodal point deregulation
120	$N_6^{(f_4)}$	$\{-f_4i'Ki, -f_4i'Kj, -f_4i'Kk\}$	$-f_4i'J$	$-f_4I$	Consumption deregulation
Block f_5: Anti-cosmic partition 5 (Dispersion regulation)					
121	$P_1^{(f_5)}$	$\{f_5iI, f_5iJ, f_5iK\}$	$f_5i'k$	f_5j	Keynesian dispersion regulation

Continued on next page...

Complete Table of the 144 Economic Pentads of Cl(6,6) – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
122	$P_2^{(f_5)}$	$\{f_5jI, f_5jJ, f_5jK\}$	$f_5i'i$	f_5k	Liberal dispersion regulation
123	$P_3^{(f_5)}$	$\{f_5kI, f_5kJ, f_5kK\}$	$f_5i'j$	f_5i	Credit dispersion regulation
124	$P_4^{(f_5)}$	$\{f_5i'Ii, f_5i'Ij, f_5i'k\}$	$f_5i'K$	f_5J	Export dispersion regulation
125	$P_5^{(f_5)}$	$\{f_5i'Ji, f_5i'Jj, f_5i'Jk\}$	$f_5i'I$	f_5K	Extraction dispersion regulation
126	$P_6^{(f_5)}$	$\{f_5i'Ki, f_5i'Kj, f_5i'Kk\}$	$f_5i'J$	f_5I	Consumption dispersion regulation
127	$N_1^{(f_5)}$	$\{-f_5iI, -f_5iJ, -f_5iK\}$	$-f_5i'k$	$-f_5j$	Dispersion deregulation
128	$N_2^{(f_5)}$	$\{-f_5jI, -f_5jJ, -f_5jK\}$	$-f_5i'i$	$-f_5k$	Dispersion deregulation
129	$N_3^{(f_5)}$	$\{-f_5kI, -f_5kJ, -f_5kK\}$	$-f_5i'j$	$-f_5i$	Dispersion deregulation
130	$N_4^{(f_5)}$	$\{-f_5i'Ii, -f_5i'Ij, -f_5i'k\}$	$-f_5i'K$	$-f_5J$	Export dispersion deregulation
131	$N_5^{(f_5)}$	$\{-f_5i'Ji, -f_5i'Jj, -f_5i'Jk\}$	$-f_5i'I$	$-f_5K$	Extraction dispersion deregulation
132	$N_6^{(f_5)}$	$\{-f_5i'Ki, -f_5i'Kj, -f_5i'Kk\}$	$-f_5i'J$	$-f_5I$	Consumption dispersion deregulation
Block f_6: Anti-cosmic partition 6 (Land regulation)					
133	$P_1^{(f_6)}$	$\{f_6iI, f_6iJ, f_6iK\}$	$f_6i'k$	f_6j	Keynesian land regulation
134	$P_2^{(f_6)}$	$\{f_6jI, f_6jJ, f_6jK\}$	$f_6i'i$	f_6k	Liberal land regulation
135	$P_3^{(f_6)}$	$\{f_6kI, f_6kJ, f_6kK\}$	$f_6i'j$	f_6i	Land credit regulation
136	$P_4^{(f_6)}$	$\{f_6i'Ii, f_6i'Ij, f_6i'k\}$	$f_6i'K$	f_6J	Export land regulation
137	$P_5^{(f_6)}$	$\{f_6i'Ji, f_6i'Jj, f_6i'Jk\}$	$f_6i'I$	f_6K	Land extraction regulation
138	$P_6^{(f_6)}$	$\{f_6i'Ki, f_6i'Kj, f_6i'Kk\}$	$f_6i'J$	f_6I	Land consumption regulation
139	$N_1^{(f_6)}$	$\{-f_6iI, -f_6iJ, -f_6iK\}$	$-f_6i'k$	$-f_6j$	Land deregulation

Continued on next page...

Complete Table of the 144 Economic Pentads of $Cl(6,6)$ – continued

ID	Pentad	Wood, Earth, Metal	Fire	Water	Economic interpretation
140	$N_2^{(f_6)}$	$\{-f_6jI, -f_6jJ, -f_6jK\}$	$-f_6i'i$	$-f_6k$	Land credit deregulation
141	$N_3^{(f_6)}$	$\{-f_6kI, -f_6kJ, -f_6kK\}$	$-f_6i'j$	$-f_6i$	Land debt deregulation
142	$N_4^{(f_6)}$	$\{-f_6i'Ii, -f_6i'Ij, -f_6i'k\}$	$-f_6i'K$	$-f_6J$	Export land deregulation
143	$N_5^{(f_6)}$	$\{-f_6i'Ji, -f_6i'Jj, -f_6i'Jk\}$	$-f_6i'I$	$-f_6K$	Land extraction deregulation
144	$N_6^{(f_6)}$	$\{-f_6i'Ki, -f_6i'Kj, -f_6i'Kk\}$	$-f_6i'J$	$-f_6I$	Land consumption deregulation

4. Comparison of Post-Nodal Point Alternatives

The following table, from the strategic analysis, compares the TRR with other possible crisis exit paths for the Chinese state.

Table 23: Comparison of post-Nodal Point alternatives

Alternative	Mass employment	Territorial anchoring	Delay	TdR compatibility	Status	Mandate of Heaven
TRR	Yes	Yes	10-15 years	High	Hybrid concept	Restoration
New physical frontiers	Yes	Yes	20-30 years	Medium	Extension of Extraction	Preserves
Polarized growth 2.0	No	No	Immediate	Low	Algorithmic Stasis	Erodes
Technological breakthrough	Yes	Yes	20-40 years	Medium	Exogenous to TdR	Neutral
Neo-feudal retreat	No	Partial	5-10 years	Null	Terminal phase	Rupture
Post-growth model	No	No	10-20 years	Null	Radical rupture	Rupture

PART C: EPISTEMOLOGY AND STRUCTURAL ISOMORPHISMS

5. Vandermeersch's Contribution: Chinese Divinatory Reason

The work of Léon Vandermeersch (*Les deux raisons de la pensée chinoise*) offers a major epistemological key. He distinguishes Western theological reason (linear causality, *causa efficiens*) from Chinese divinatory reason (structural correlation).

Western models (DSGE) fail because they project a causal logic onto a system that obeys a logic of correlations (Yi Jing, Wuxing). Vandermeersch demonstrates that Chinese writing was not born from the transcription of speech, but from the recording of divinatory operations: $XY \times N : M \rightarrow P$.

The oracle bone inscriptions (jiaguwen, 甲骨文, c. 1200-1050 BC) are the key evidence. These inscriptions, carved on turtle plastrons and ox scapulae, are not phonetic transcriptions but formalizations of relational structures. The diviner would pose a binary question (e.g., “Will the king’s hunt be successful? Yes/No”), then interpret the crack patterns produced by heat application. This process generated a formal system of 64 hexagrams (6 binary traits), each representing a stable configuration of forces.

The Cl(6,6) spectral dashboard is the modern equivalent of this state divination: it does not seek causes, but stable configurations — just as the diviner did not seek the cause of a phenomenon but its place in the total correlation of the world.

This epistemological distinction is fundamental: the West seeks the *why* (the cause), China seeks the *how* (the correlation). The Cl(6,6) model aligns with the latter: it maps the space of possible economic configurations without postulating a linear causality.

6. Philosophical Synthesis: The spectral resolution of the Xiong Shili – Nāgārjuna controversy

The topological model brings a mathematical resolution to a historical tension of Asian thought:

Xiong Shili (Neo-Confucian, 1885-1968): Defends harmony ($d = 5$), a creative balance between unity and diversity. For Xiong, reality is the constant alternation between the One and the Many, between the immutable substance (*ti* 體) and its functions (*yong* 用). This is the optimal spectral regime of tempered growth, where the system is sufficiently diversified to be resilient but sufficiently coherent to be governed. Xiong’s philosophy, developed in his *Xin weishi lun* (New Treatise on Consciousness-Only, 1932), posits that the substance is not static but constantly transforming itself into functions — exactly as the Cl(6,6) model describes the transition between attractors through the Dirac operator.

Nāgārjuna (Madhyamaka, c. 150-250 CE): Defends Emptiness (*śūnyatā*, $d \rightarrow 0$), the dissolution of forms, condition of possibility of change. For Nāgārjuna, all phenomena are empty of intrinsic existence (*svabhava*); they exist only by their relations (*pratītyasamutpāda*). This is the condition of creative destruction necessary at the Nodal Point to reset the system. The total dissolution of forms ($d = 0$) is the moment of pure potentiality, the “cloud of possibilities” from which a new configuration can emerge. Nāgārjuna’s *Mūlamadhyamakakārikā* (Fundamental Verses on the Middle Way) demonstrates that emptiness is not nihilism but the condition of all arising — exactly as the spectral gap ($\text{gap} \rightarrow 0$) is the condition for a phase transition.

The TRR is the political operator that allows the system to traverse Emptiness ($d \rightarrow 0$) without getting lost there (retropolarity), to regain Harmony ($d = 5$) via the threshold N_4 . It is the institutional translation of the Madhyamaka dialectic: the destruction of obsolete forms (ghost cities, atrophied countryside) and the recreation of new forms (regenerated spaces, green jobs) in a controlled process. The TRR avoids both the sclerosis of fixed forms (Xiong’s risk) and the nihilism of pure dissolution (Nāgārjuna’s risk).

7. The Aporias of Neoclassical and Marxist Economic Theories

The Cl(6,6) model escapes the aporias that paralyze Western economic theories:

The neoclassical aporia (Rebour): The model of pure and perfect competition (Convergence Theory) is not capable of accounting for spatial agglomeration. Yet the city has structured the landscape for at least 5000 years. If it were true, economic and geographic space would be homogeneous and amorphous. TdR resolves this by modeling growth as a polarized disequilibrium. The neoclassical model cannot explain why the economy tends towards concentration rather than dispersion. The Cl(6,6) model integrates this by making dispersion a binary variable that switches from 0 (concentration) to 1 (dispersion) according to the phase of the cycle. This is exactly what the “Dispersion” variable in the 64-configuration space captures: the transition from a concentrated regime (urbanization, frontier closure) to a fragmented regime (ghost cities, rural exodus).

The Marxist aporia (Lavigne): The labor theory of value is tautological (the value of labor is defined by labor). Marx cannot explain why some labors are more “productive” than others in value creation, except by invoking a metaphysical distinction between concrete labor and abstract labor. The Cl(6,6) model defines value not as a substance, but as a topological relationship between a stock (Space of Value) and a flow (Space of Wealth). Value is not an intrinsic property of the commodity; it is the product of a transfer operation from a latent state to a monetized state. This is exactly what the Extraction mechanism formalizes: the transfer of land (latent value) to the productive circuit through anticipatory credit (monetized wealth). The “Extraction” variable in the 64-configuration space captures this: 0 = Nodal point (no transfer possible), 1 = active extraction (transfer in progress).

8. Isomorphism with Tang Alliances (Wufu) and Mandarin Phonology

The universality of the invariant $64 \rightarrow 20$ is verified in the analysis of the kinship systems of the Tang dynasty (Wufu) and in Mandarin phonology.

Tang Alliances (Wufu, 五服): The 64 matrimonial configurations (including prohibitions) reduce to 20 stable alliance attractors, filtered by pentads (the 5 degrees of mourning). The Tang aristocracy practiced an extremely complex system of matrimonial alliances, governed by five degrees of mourning (*wufu*) which defined the circle of permitted and prohibited unions: - *Zhan cui* (斬衰, 3 years): father, husband, eldest son - *Zi cui* (齊衰, 1-3 years): mother, grandparents, uncles - *Da gong* (大功, 9 months): married sisters, cousins - *Xiao gong* (小功, 5 months): great-uncles, first cousins once removed - *Si ma* (緦麻, 3 months): distant relatives

This system, analyzed through the prism of Clifford algebras, reveals 64 possible configurations that reduce to 20 stable alliance attractors — exactly the same number as the 20 economic attractors. This is not a coincidence, but the expression of a universal structural law of complex systems.

Mandarin phonology: The 64 possible combinations (24 initials + 40 finals) reduce to 20 phonological classes effectively used, filtered by articulatory constraints. Mandarin Chinese has 24 possible initials and 40 possible finals (if one counts tones), which gives 960 theoretical combinations. But the phonological system actually uses only a subset; by decomposing into 6 binary articulatory features (voicing, place of articulation, manner of articulation, aspiration, tone, rounding), one obtains $2^6 = 64$ possible configurations, which reduce to 20 effectively used phonological classes. The constraints are articulatory (the tongue cannot be in two positions at once, the vocal cords cannot be open and closed simultaneously), exactly as in economics the constraints are institutional and physical (one cannot have both high inflation and production collapse).

The economy thus obeys the same universal law of structuring of complex systems as biology (genetic code: 64 codons $\rightarrow$ 20 amino acids), particle physics (64 spin configurations $\rightarrow$ 20 stable states), kinship systems (Wufu), or phonology (Mandarin). This universal invariant $64 \rightarrow 20$ is not a metaphor; it is the mathematical signature of the topology of complex systems.